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1 Supplementary Material

1.1 Scope

This supplement records the equations, normalizations, numerical definitions, and validation statements required to reproduce the manuscript. It does not change the C22 scientific freeze.

2 Charge and operator audit

The charge of a monomial is the sum of field charges, with conjugated fields carrying the additive inverse charge. The enumerator lists all commuting scalar monomials through engineering dimension four, groups Hermitian-conjugate pairs, and compares invariance under the projected generator P with invariance under both P and K . The resulting comparator-only classes are the cross kinetic term, the bilinear $A^\dagger B$, two mixed cubics, two modulus-weighted bilinears, $(A^\dagger B)^2$, and the off-diagonal Higgs portal. Machine output is stored in audits/dual_tp_operator_basis.json.

3 Vacuum conditions

For the electroweak and two-dark-field quartic,

$$V_4 = \lambda_H x^2 + \lambda_A y^2 + \lambda_B z^2 + \lambda_{HA} xy + \lambda_{HB} xz + \lambda_{AB} yz,$$

exact boundedness is copositivity of the symmetric matrix defined in the main text. The implementation evaluates the full three-variable copositivity conditions, then separately checks all stationary points of the phase-minimized tree potential. The analytic positive-portal certificate is used only as a sufficient shortcut and is never described as necessary.

4 Scalar partial-wave conventions

Two-particle states contain the conventional $1/\sqrt{2}$ factor for identical pairs. With

$$\mathcal{M}(\cos \theta) = 16\pi \sum_J (2J+1) a_J P_J(\cos \theta),$$

the projection is

$$a_J = \frac{1}{32\pi} \int_{-1}^1 d\cos \theta P_J(\cos \theta) \mathcal{M}(\cos \theta),$$

with the corresponding Wigner- d generalization for helicity amplitudes. The audit applies $|\text{Re } a_J| \leq 1/2$ to properly normalized nonsingular physical channels.

5 Sequential kinematics derivation

In the center-of-mass frame of $X(M)S(m) \rightarrow X(M)S(m)$,

$$u = M^2 + m^2 - 2E_X E_S - 2p^2 \cos \theta.$$

Setting the exchanged scalar on shell gives

$$M^2 - 2E_X E_S - 2p^2 \cos \theta = 0.$$

The first and last contact with the physical angular domain occur at $\cos \theta = -1$ and $+1$, yielding

$$\sqrt{s}_{\text{low}} = \sqrt{2M^2 + m^2}, \quad \sqrt{s}_{\text{high}} = \frac{M^2 - m^2}{m}.$$

Squaring the ordering condition shows that $\sqrt{s}_{\text{high}} > \sqrt{s}_{\text{low}}$ exactly when $M > 2m$. Thus the pole interval and the physical decay $X \rightarrow SS^\dagger$ are the same kinematic fact.

6 One-loop running

The low-energy scalar quartic and dimensionful beta functions were obtained independently from the effective-potential Hessian and tensor contractions. The parent calculation evolves the Standard Model gauge and third-family Yukawa couplings, g_m , g_E , and all 17 CP-conserving parent scalar quartics on the protected zero-mixing slice. The numerical matching is a single step at 1 TeV. This approximation is the reason the reported high scales are model-dependent estimates rather than precision thresholds.

7 Thermal system

For each complex scalar, particle and antiparticle are assumed symmetric. The equilibrium yield per charge state is evaluated with the modified Bessel function K_2 . The finite-width Higgs-mediated rate is inserted into the Gondolo-Gelmini integral. The coupled collision polynomials were separately tested for detailed balance, and conversion was verified to conserve total dark particle number.

The final EOS model uses ideal-gas massive thresholds and a smooth QCD parton-to-hadron interpolation. A plus-or-minus five percent variation of the thermodynamic functions yields the quoted relic envelope. This variation is a sensitivity diagnostic, not a substitute for a dedicated lattice-QCD equation of state.

8 Direct detection normalization

For a complex scalar with portal $\lambda_{HS}(H^\dagger H)|S|^2$, the per-nucleon scalar cross section used in the audit is

$$\sigma_N^{SI} = \frac{\lambda_{HS}^2 f_N^2 \mu_{NS}^2 m_N^2}{4\pi m_h^4 m_S^2},$$

with $f_N = 0.30$ and $\mu_{NS} = m_N m_S / (m_N + m_S)$. The experiment comparison is made only after multiplying by $\xi_i = \Omega_i / \Omega_{DM}$. The two pole spectra are summed because their recoil-scale difference is below 0.5 percent.

9 Validation inventory

The final suite contains 34 passing tests. It covers operator counts, analytic spectra, randomized mass and unitarity checks, copositivity and vacuum conditions, one-loop scalar projections, parent gauge coefficients, detailed balance, relic cross-checks, LZ interpolation, electroweak Goldstone identities, parent Ward contractions, cubic-rate suppression, metastability cutoff ordering, prompt parent decays, and the explicit unscored sequential interval.

10 Machine-readable records

The reproducibility archive includes the JSON records for every promoted numerical statement, the Python scripts that generated them, the dependency ledger, the v0.3 theory definition, checkpoints C1-C22 where available, and a SHA-256 manifest.

11 Frozen validation dossier

11.1 S1 Ternary Polarity Core-Survival Gate

11.2 Scope and decision

This report tests the present TP-1 scalar-gauge core before book chapters are drafted around it. TP-1 consists of the Standard Model, a hidden compact $U(1)_X$, a charge-three complex scalar S , a charge-one complex scalar χ , and the residual Z_3 left after $\langle S \rangle \neq 0$. With $Q_X(S) = 3$ and $Q_X(\chi) = 1$, this is the same local- Z_3 model class studied by Ko and Tang, not a novel model class by itself.¹

Four conclusions can now be stated.

1. The full tree-level boundedness problem admits an exact reduction to a one-dimensional positivity test.
2. Every generic radial stationary branch can be enumerated algebraically through at most a cubic equation.
3. A healthy inert benchmark exists and the branch enumerator passed an independent randomized numerical validation.
4. The dual labels (q_m, q_E) have no additional physics if all operators and observables depend only on $q_P = q_m + q_E$. A genuinely distinct theory must retain the full $Z_3 \times Z_3$ selection rule or another observable action of its kernel.

The high-energy scalar unitarity matrix and the scalar-only part of the one-loop quartic RG system have also been generated reproducibly. Finite-energy unitarity, the gauge/Yukawa completion of all beta functions, loop-improved vacuum stability, and phenomenology remain open. TP-1 therefore survives this checkpoint as a consistent candidate EFT branch, not as a completed theory of nature.

MODEL-DEPENDENT RESULT

A nonempty, explicitly checked tree-level parameter region has the required $v \neq 0$, $f \neq 0$, $\langle \chi \rangle = 0$, positive radial masses, bounded potential, and perturbative high-energy scalar amplitudes.

OPEN PROBLEM

No unique empirical prediction of Ternary Polarity has yet been established. The strongest current route is a kernel-sensitive selection rule in a true $Z_3 \times Z_3$ realization.

11.3 1. Canonical scalar sector

Use natural units and write

$$H^\dagger H = \frac{h^2}{2}, \quad S = \frac{s}{\sqrt{2}} e^{i\alpha}, \quad \chi = \frac{x}{\sqrt{2}} e^{i\beta}, \quad h, s, x \geq 0.$$

The most general renormalizable potential for the stated fields and gauge charges is

$$\begin{aligned} V = & -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2 - \mu_S^2 S^\dagger S + \lambda_S (S^\dagger S)^2 + \mu_\chi^2 \chi^\dagger \chi + \lambda_\chi (\chi^\dagger \chi)^2 \\ & + \lambda_{HS} (H^\dagger H) (S^\dagger S) + \lambda_{H\chi} (H^\dagger H) (\chi^\dagger \chi) + \lambda_{S\chi} (S^\dagger S) (\chi^\dagger \chi) + (\kappa S^\dagger \chi^3 + \text{h.c.}). \end{aligned}$$

A field redefinition makes $\kappa \geq 0$. For $sx \neq 0$, the phase minimizes at

$$\cos(3\beta - \alpha) = -1.$$

The phase-minimized radial potential is therefore

$$\begin{aligned} V(h, s, x) = & -\frac{1}{2}\mu_H^2 h^2 + \frac{1}{4}\lambda_H h^4 - \frac{1}{2}\mu_S^2 s^2 + \frac{1}{4}\lambda_S s^4 + \frac{1}{2}\mu_\chi^2 x^2 + \frac{1}{4}\lambda_\chi x^4 \\ & + \frac{1}{4}\lambda_{HS} h^2 s^2 + \frac{1}{4}\lambda_{H\chi} h^2 x^2 + \frac{1}{4}\lambda_{S\chi} s^2 x^2 - \frac{1}{2}\kappa s x^3. \end{aligned}$$

The κ operator is the gauge-invariant parent of the low-energy Z_3 cubic interaction. Its normalization agrees with the local- Z_3 construction after the corresponding charge and field rescaling.¹ Global- Z_3 singlet models contain the analogous symmetry-breaking cubic after the parent radial mode is removed.²

11.4 2. Exact boundedness reduction

The quartic part satisfies

$$4V_4 = \lambda_H h^4 + \lambda_S s^4 + \lambda_\chi x^4 + \lambda_{HS} h^2 s^2 + \lambda_{H\chi} h^2 x^2 + \lambda_{S\chi} s^2 x^2 - 2\kappa s x^3.$$

The $x = 0$ boundary is bounded strictly upward when

$$\lambda_H > 0, \quad \lambda_S > 0, \quad \lambda_{HS} + 2\sqrt{\lambda_H \lambda_S} > 0.$$

For $x > 0$, define $u = h/x$ and $t = s/x$. Then

$$\frac{4V_4}{x^4} = \lambda_H u^4 + B(t)u^2 + P(t),$$

where

$$B(t) = \lambda_{HS} t^2 + \lambda_{H\chi}, \quad P(t) = \lambda_S t^4 + \lambda_{S\chi} t^2 - 2\kappa t + \lambda_\chi.$$

For fixed t , set $z = u^2 \geq 0$. The expression is a quadratic in z , and its exact minimum is

$$R(t) = \begin{cases} P(t), & B(t) \geq 0, \\ P(t) - \frac{B(t)^2}{4\lambda_H}, & B(t) < 0. \end{cases}$$

Consequently, apart from non-strict flat-direction boundaries, the full radial quartic is bounded from below if and only if the $x = 0$ condition holds and

$$R(t) > 0 \quad \text{for every } t \geq 0.$$

This is stronger than the earlier sufficient condition obtained by assuming all portals nonnegative and discarding positive terms. It is also practical: $R(t)$ is piecewise quartic and can be certified by checking endpoints, switching points $B(t) = 0$, and real nonnegative stationary points in each piece.

DERIVED RESULT

The three-field boundedness problem has been reduced exactly to a one-variable positivity problem. This is a mathematical result for TP-1's stated potential, not an experimental claim.

11.5 3. Complete generic stationary-branch enumeration

The radial stationarity equations are

$$0 = h \left[-\mu_H^2 + \lambda_H h^2 + \frac{1}{2} \lambda_{HS} s^2 + \frac{1}{2} \lambda_{H\chi} x^2 \right],$$

$$0 = -\mu_S^2 s + \lambda_S s^3 + \frac{1}{2} \lambda_{HS} h^2 s + \frac{1}{2} \lambda_{S\chi} s x^2 - \frac{1}{2} \kappa x^3,$$

$$0 = x \left[\mu_\chi^2 + \lambda_\chi x^2 + \frac{1}{2} \lambda_{H\chi} h^2 + \frac{1}{2} \lambda_{S\chi} s^2 - \frac{3}{2} \kappa s x \right].$$

For $x = 0$, the candidates are the origin, the H -only extremum, the S -only extremum, and the inert HS extremum. Define

$$\Delta = \lambda_H \lambda_S - \frac{1}{4} \lambda_{HS}^2.$$

The inert solution is

$$v^2 = \frac{\lambda_S \mu_H^2 - \frac{1}{2} \lambda_{HS} \mu_S^2}{\Delta}, \quad f^2 = \frac{\lambda_H \mu_S^2 - \frac{1}{2} \lambda_{HS} \mu_H^2}{\Delta},$$

when both right-hand sides are positive. Its energy is

$$V_{HS} = -\frac{1}{4} (\mu_H^2 v^2 + \mu_S^2 f^2).$$

Its CP-even radial mass matrix is

$$\mathcal{M}_{hs}^2 = \begin{pmatrix} 2\lambda_H v^2 & \lambda_{HS} v f \\ \lambda_{HS} v f & 2\lambda_S f^2 \end{pmatrix},$$

while the two real components of the inert complex scalar have common mass

$$m_\chi^2 = \mu_\chi^2 + \frac{1}{2} \lambda_{H\chi} v^2 + \frac{1}{2} \lambda_{S\chi} f^2.$$

For $\kappa > 0$, any stationary point with $x > 0$ must also have $s > 0$. This removes an apparent branch: $s = 0, x > 0$ is inconsistent because the s equation contains $-\kappa x^3/2$.

Both remaining mixed cases reduce to the same cubic. For a branch with $h > 0$, analytically eliminate h and define Schur-complement parameters

$$\begin{aligned} \bar{\mu}_S^2 &= \mu_S^2 - \frac{\lambda_{HS} \mu_H^2}{2\lambda_H}, & \bar{\mu}_\chi^2 &= \mu_\chi^2 + \frac{\lambda_{H\chi} \mu_H^2}{2\lambda_H}, \\ \bar{\lambda}_S &= \lambda_S - \frac{\lambda_{HS}^2}{4\lambda_H}, & \bar{\lambda}_\chi &= \lambda_\chi - \frac{\lambda_{H\chi}^2}{4\lambda_H}, & \bar{\lambda}_{S\chi} &= \lambda_{S\chi} - \frac{\lambda_{HS} \lambda_{H\chi}}{2\lambda_H}. \end{aligned}$$

For a branch with $h = 0$, remove the bars by using the original parameters. With $t = s/x > 0$, every nondegenerate mixed stationary point obeys

$$\begin{aligned} 0 &= \left(\frac{1}{2} \mu_S^2 \lambda_{S\chi} + \mu_\chi^2 \lambda_S \right) t^3 - \frac{3}{2} \kappa \mu_S^2 t^2 \\ &\quad + \left(\mu_S^2 \lambda_\chi + \frac{1}{2} \mu_\chi^2 \lambda_{S\chi} \right) t - \frac{1}{2} \kappa \mu_\chi^2, \end{aligned}$$

using either the barred or unbarred set as just described. Each positive real root is retained only when it gives $x^2 > 0$, $s = tx > 0$, and, for the active-Higgs branch, $h^2 > 0$. The Hessian and vacuum energy are then evaluated directly. Homogeneity supplies the useful stationary-point check

$$V_{stat} = \frac{1}{4} (-\mu_H^2 h^2 - \mu_S^2 s^2 + \mu_\chi^2 x^2).$$

This procedure enumerates every generic radial stationary point. Measure-zero cases such as $\kappa = 0$, vanishing quartics, or singular Schur complements must be handled separately because they can introduce enhanced symmetries or continuous stationary sets.

11.6 4. Numerical certification

An illustrative benchmark was defined by

$$v = 246 \text{ GeV}, \quad f = 1000 \text{ GeV}, \quad m_\chi = 500 \text{ GeV},$$

$$(\lambda_H, \lambda_S, \lambda_\chi) = (0.13, 0.30, 0.25),$$

$$(\lambda_{HS}, \lambda_{H\chi}, \lambda_{S\chi}, \kappa) = (0.02, 0.05, 0.08, 0.05).$$

The mass parameters were fixed by the inert stationary conditions. The exact radial boundedness test had a positive minimum

$$\min_{t \geq 0} R(t) = 0.2292548748 \quad \text{at} \quad t = 0.3372682429.$$

The inert stationary point was the lowest enumerated candidate, with

$$V_{HS} = -7.5421601053 \times 10^{10} \text{ GeV}^4.$$

The three radial Hessian eigenvalues were

$$(1.56927 \times 10^4, 2.50000 \times 10^5, 6.00041 \times 10^5) \text{ GeV}^2.$$

An independent differential-evolution search followed by bounded local minimization returned $(h, s, x) = (246.0000024, 999.9999998, 0) \text{ GeV}$ with the same energy to the displayed accuracy.

The branch enumerator was then tested on 40 randomly generated bounded potentials using a fixed seed. The direct global optimizer and analytic enumeration agreed in all 40 cases under a relative energy tolerance of 2×10^{-6} . The winning branches included the origin, H -only, S -only, inert HS , mixed $S\chi$, and fully mixed $HS\chi$ vacua. This is a numerical cross-check, not a proof of the optimizer; the algebraic enumeration supplies the completeness argument in generic cases.

11.7 5. High-energy scalar unitarity

The finite-energy convention used in modern Z_3 singlet analyses constructs the $J = 0$ matrix with explicit identical-particle factors and imposes

$$|\text{Re } a_0^i| \leq \frac{1}{2}.$$

Exchange diagrams can make finite-energy constraints stronger than the infinite-energy quartic limit, especially when cubic interactions are important. TP-1 must therefore retain a finite-energy scan in its final audit.

As a first exact layer, all eight real scalar components were included: four from H , two from S , and two from χ . There are 36 unordered two-particle channels. If

$$V_4 = \frac{1}{4!} \lambda_{abcd} \phi_a \phi_b \phi_c \phi_d,$$

the normalized high-energy matrix is

$$L_{(ij)(kl)} = \frac{\lambda_{ijkl}}{\sqrt{2^{\delta_{ij}} 2^{\delta_{kl}}}}, \quad a_0 = -\frac{L}{16\pi}.$$

Thus every eigenvalue L_i must satisfy $|L_i| \leq 8\pi$. For the benchmark above, the spectral radius was 1.2337507508, giving

$$\max_i |a_0^i| = 0.0245447.$$

The benchmark therefore passes the scalar-quartic high-energy test by a wide margin. One-coupling-at-a-time checks reproduce the expected $O(4)$ eigenvalue $6\lambda_H$ and $O(2)$ eigenvalues $4\lambda_S$ and $4\lambda_\chi$, providing a normalization check.

PARTIAL RESULT

The complete 36-channel high-energy scalar quartic matrix has been generated and checked. This does not yet replace the required finite-energy calculation with scalar exchange, gauge/Goldstone channels, pole excision, and an explicit energy range.

11.8 6. One-loop RGE checkpoint

For real scalars in the convention above, the scalar-only one-loop tensor equation is

$$(16\pi^2)\beta_{abcd} = \lambda_{abef}\lambda_{efcd} + \lambda_{acef}\lambda_{efbd} + \lambda_{adef}\lambda_{efbc}.$$

An exact-arithmetic tensor generator gives

$$(16\pi^2)\beta_{\lambda_H}^{scalar} = 24\lambda_H^2 + \lambda_{HS}^2 + \lambda_{H\chi}^2,$$

$$(16\pi^2)\beta_{\lambda_S}^{scalar} = 20\lambda_S^2 + 2\lambda_{HS}^2 + \lambda_{S\chi}^2,$$

$$(16\pi^2)\beta_{\lambda_\chi}^{scalar} = 20\lambda_\chi^2 + 2\lambda_{H\chi}^2 + \lambda_{S\chi}^2 + 18\kappa^2,$$

$$(16\pi^2)\beta_{\lambda_{HS}}^{scalar} = 12\lambda_H\lambda_{HS} + 8\lambda_S\lambda_{HS} + 4\lambda_{HS}^2 + 2\lambda_{H\chi}\lambda_{S\chi},$$

$$(16\pi^2)\beta_{\lambda_{H\chi}}^{scalar} = 12\lambda_H\lambda_{H\chi} + 8\lambda_\chi\lambda_{H\chi} + 4\lambda_{H\chi}^2 + 2\lambda_{HS}\lambda_{S\chi},$$

$$(16\pi^2)\beta_{\lambda_{S\chi}}^{scalar} = 8\lambda_S\lambda_{S\chi} + 8\lambda_\chi\lambda_{S\chi} + 4\lambda_{S\chi}^2 + 4\lambda_{HS}\lambda_{H\chi} + 36\kappa^2,$$

$$(16\pi^2)\beta_{\kappa}^{scalar} = \kappa(12\lambda_\chi + 6\lambda_{S\chi}).$$

The generator reproduces independently known $O(4)$ and $O(2)$ scalar coefficients. These equations remain a checkpoint until a second implementation verifies every mixed and κ -dependent coefficient.

For the minimal hidden field content, the one-loop Abelian gauge coefficient is fixed by the two complex scalars:

$$(16\pi^2)\beta_{g_X} = \frac{10}{3}g_X^3,$$

because $\frac{1}{3}(3^2+1^2) = 10/3$. A general multiple- $U(1)$ treatment must use a gauge-coupling matrix rather than treating kinetic mixing as an afterthought.^{4,5} If no field carries both hypercharge and X charge, the inhomogeneous one-loop source proportional to $\sum_i Y_i Q_{X,i}$ vanishes. A nonzero UV threshold or added bi-charged matter can nevertheless generate kinetic mixing, so $\epsilon = 0$ is a model boundary condition, not a consequence of residual Z_3 alone.

The complete one-loop system still requires Standard Model gauge and Yukawa terms, $U(1)_X$ gauge terms in every scalar beta function, masses, scalar anomalous dimensions, threshold matching at f , and optional nonminimal gravitational couplings. No RG-stability scale is claimed in this report.

11.9 7. The dual-charge distinctness theorem

Let

$$G = Z_3 \times Z_3, \quad \pi(a, b) = a + b \pmod{3}, \quad K = \ker \pi = \{(0, 0), (1, 2), (2, 1)\}.$$

Suppose every field transformation, allowed operator, Hamiltonian term, state label, and observable depends on (a, b) only through $\pi(a, b)$. Then the action of K is physically trivial and the representation factors through

$$G/K \simeq Z_3.$$

Any two assignments in the same fibre of π are operationally indistinguishable. The theory is therefore equivalent, with respect to all specified observables, to an ordinary Z_3 theory plus redundant labels or unresolved multiplicity.

DERIVED RESULT

The fibre cardinality of three does not by itself create three physical states, three generations, or new selection rules. Physical distinctness requires a kernel-sensitive operation or observable.

Structure	Operator condition	Physical content
Quotient-only H_0	$\sum_i (a_i + b_i) = 0 \pmod{3}$	Ordinary Z_3
Full dual symmetry H_1	$\sum_i a_i = 0$ and $\sum_i b_i = 0 \pmod{3}$	Strictly stronger selection rules

Take two complex scalars

$$A \sim (1, 0), \quad B \sim (0, 1).$$

Both have quotient charge $q_P = 1$. Under quotient-only Z_3 , the bilinear $A^\dagger B$ is neutral and allowed. Under the full product symmetry it carries $(2, 1) \neq (0, 0)$ and is forbidden. The cubic operators $A^2 B$ and AB^2 are likewise quotient-neutral but forbidden by $Z_3 \times Z_3$.

The lowest-dimension difference is therefore already dimension two:

$$A^\dagger B: \quad H_0 \text{ allows,} \quad H_1 \text{ forbids.}$$

With a neutral scalar mediator R , $RA^\dagger B$ supplies a dimension-three decay or conversion vertex in H_0 but remains forbidden in H_1 . Hence an exact full symmetry predicts

$$\Gamma(A \rightarrow B + R) = 0$$

whenever that decay would rely on this operator and no symmetry-breaking spurion is present. Observation of such mixing or conversion would falsify the exact H_1 assignment. Non-observation would not by itself establish Ternary Polarity, because an ordinary model can set the same coupling to zero for another reason.

A minimal anomaly-free parent uses only scalars beyond the SM:

$$U(1)_m \times U(1)_E \xrightarrow{\langle S_m \rangle, \langle S_E \rangle} Z_3^{(m)} \times Z_3^{(E)},$$

with $S_m \sim (3, 0)$, $S_E \sim (0, 3)$, $A \sim (1, 0)$, and $B \sim (0, 1)$. Because no new chiral fermions are required, gauge anomalies are absent in this minimal scalar realization. Its two string species give a kernel-sensitive holonomy matrix:

Probe	m -string phase	E -string phase
$A \sim (1, 0)$	ω	1

Probe	m -string phase	E -string phase
$B \sim (0, 1)$	1	ω

This is distinct in principle from a single residual Z_3 , whose probes of equal quotient charge have the same discrete holonomy. Discrete gauge symmetries and their Aharonov–Bohm observables are established structures, so the novelty could only lie in a constrained physical identification and its consequences, not in the existence of such holonomies.⁶

The cost is significant: a second gauge factor, gauge boson, breaking scalar, portals, kinetic-mixing parameters, and defect sector are introduced. The words “mass” and “energy” do not acquire physical meaning from the group labels. Until observables link q_m and q_E to independently measured mass- and energy-related quantities, they should be described technically as two internal ternary charges.

11.10 8. Hostile-referee assessment

Test	Status	Reason
Lorentz-covariant local action	Achieved	Standard gauge-scalar EFT
Healthy tree-level kinetic terms	Achieved	Canonical signs
Complete renormalizable scalar potential	Achieved	All allowed operators retained
Exact tree-level boundedness test	Achieved	One-variable reduction
Generic tree-level branch enumeration	Achieved	Cubic reduction plus boundaries
Example global inert vacuum	Achieved	Algebraic and numerical agreement
High-energy scalar unitarity	Achieved for benchmark	36-channel quartic matrix
Finite-energy unitarity	Open	Exchange and gauge channels pending
Complete one-loop RG	Partial	Scalar-only tensor and g_X beta derived
Loop-improved global vacuum	Open	Scheme and counterterms not frozen
Distinctness from ordinary Z_3	Failed for TP-1	Present dynamics use only one residual charge
Distinct dual-charge model	Partial	Exact selection-rule candidate, no fitted benchmark
Unique empirical prediction	Open	Null conversion rule is conditional
Three-generation explanation	Not achieved	Fibre multiplicity is not a mechanism
Gravity-scale prediction	Not achieved	Matching relation still contains free inputs

The decisive scientific conclusion is narrow but useful: TP-1 is a viable local- Z_3 EFT candidate in an established model class. Its current Ternary Polarity identity resides in interpretation, not an empirical distinction. TP-1D, the full dual-charge extension, supplies genuinely stronger selection rules but pays a substantial complexity cost and has not yet demonstrated parameter compression.

11.11 9. Next mandatory calculations

1. Generate the full finite-energy $2 \rightarrow 2$ scalar/Goldstone matrix, excise pole regions consistently, and scan the relevant center-of-mass energy range.
2. Independently regenerate the complete one-loop RGEs with a second implementation, including all gauge, Yukawa, kinetic-mixing, mass, and threshold terms.
3. Turn the $A^\dagger B$ selection-rule difference into a minimal production-and-decay benchmark with fewer free parameters than measured channels.

4. Compare the dual model against an ordinary Z_3 two-species model using explicit parameter counts and likelihood-independent falsification criteria.
5. Only after these steps, select the canonical branch for dependency-safe manuscript drafting.

11.12 Reproducibility record

The checkpoint uses Python 3 with NumPy 2.3.5 and SciPy 1.17.0. The random seed is 20260915. The included scripts are:

- `vacuum_audit.py`: exact branch reduction, benchmark evaluation, and direct global minimization;
- `test_vacuum_enumerator.py`: 40-case randomized cross-check;
- `scalar_rge_generator.py`: exact-arithmetic real-scalar tensor contraction;
- `unitarity_matrix.py`: 36-channel high-energy scalar matrix.

Machine-readable outputs record the benchmark, stationary points, optimizer residuals, random-validation result, RGE coefficients, and unitarity spectrum. The calculations use natural units; dimensionful scalar inputs are in GeV or GeV^2 .

11.13 Sources

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11.14 S2 Ternary Polarity Finite-Energy Unitarity Audit

11.15 Result

The TP-1 benchmark passes the complete tree-level **physical-scalar** finite-energy unitarity calculation over $250.54 \text{ GeV} \leq \sqrt{s} \leq 20 \text{ TeV}$, after explicit open-channel selection and pole exclusions. It also passes the 36-channel scalar-plus-Goldstone calculation used as a gaugeless proxy for longitudinal-vector scattering.

This does not yet constitute a full gauge-theory unitarity proof. Gauge vertices, transverse-vector channels, ghosts where required by gauge fixing, and a second implementation remain outside the present calculation.

11.16 1. Partial-wave convention

For two-particle channels $a = \{1, 2\}$ and $b = \{3, 4\}$,

$$a_0^{ba} = \frac{1}{32\pi} \sqrt{\frac{4|\mathbf{p}_b||\mathbf{p}_a|}{2^{\delta_{12}}2^{\delta_{34}}s}} \int_{-1}^1 d(\cos \theta) \mathcal{M}_{ba}(\cos \theta).$$

The tree-level criterion is

$$|\text{Re } a_0^i| \leq \frac{1}{2}.$$

A more conservative perturbative-trust diagnostic is $|a_0^i| \leq 1/6$. The calculation follows the normalization and pole logic of the general SARAH scalar-unitarity treatment and its Z_3 application.^{1,2}

11.17 2. Amplitude construction

The eight real scalar components are rotated to the mass basis. At the inert benchmark the tree-level squared masses are

$$(0, 0, 0, 0, 15692.7325, 250000, 250000, 600041.4275) \text{ GeV}^2.$$

The four massive states have masses

$$(125.2706, 500, 500, 774.6234) \text{ GeV}.$$

For cubic tensors g_{ijk} and quartic tensors λ_{ijkl} , every matrix element includes

$$\mathcal{M}_{ab \rightarrow cd} = -\lambda_{abcd} - \sum_e \frac{g_{abe}g_{cde}}{s - m_e^2} - \sum_e \frac{g_{ace}g_{bde}}{t - m_e^2} - \sum_e \frac{g_{ade}g_{bce}}{u - m_e^2}.$$

Both numerical Gauss–Legendre integration and analytic logarithmic integration of the t - and u -channel propagators were implemented.

11.18 3. Independent validation

The engine was reduced to the global- Z_3 singlet model with mass M , self-coupling λ , and cubic μ_3 . Its three-channel numerical eigenvalues were compared with Eqs. (3.6)–(3.7) of Hektor, Hryczuk, and Kannike.²

$$a_0^{(1)} = -\frac{\sqrt{s(s-4M^2)} [4\lambda(s-M^2) + 9\mu_3^2]}{32\pi s(s-M^2)},$$

$$a_0^{(2)} = \frac{4\lambda(4M^2-s) + 9\mu_3^2 \ln[(s-3M^2)/M^2]}{16\pi\sqrt{s(s-4M^2)}}.$$

At $s/M^2 = 4.05, 5, 7.5, 12, 50$, the maximum relative discrepancy was

$$1.23 \times 10^{-15}.$$

This validates the amplitude signs, identical-state normalization, phase-space prefactor, exchange terms, and angular integration against an independent published result.

11.19 4. Pole policy

Tree-level amplitudes are not reliable on resonances. The scan uses the Goodsell–Staub s -channel safety condition

$$\left| 1 - \frac{s}{m_e^2} \right| > C_S, \quad C_S = 0.25.$$

Sixteen grid points were removed by this cut. For a real t - or u -channel pole inside the angular interval, every two-particle state participating in the offending matrix element was removed before diagonalization. This is intentionally conservative and may discard more information than a block-by-block SARAH implementation.

The numerical pole cuts are safety prescriptions, not fundamental physical bounds. Earlier trilinear-unitarity work likewise emphasizes that Born amplitudes cannot be trusted on resonances.³ A future loop-level resonance calculation would require widths and a consistent resummation scheme.

11.20 5. Physical-scalar scan

Ten unordered channels were constructed from the four massive scalar states. The energy grid included 260 logarithmic points plus points immediately above all six distinct two-particle thresholds:

$$250.5413, 625.2706, 899.8940, 1000, 1274.6234, 1549.2468 \text{ GeV}.$$

After pole exclusions, 273 energy points were evaluated. The largest eigenvalue magnitude was

$$\max_{\sqrt{s}} \max_i |a_0^i| = 0.02027154$$

at the scan endpoint $\sqrt{s} = 20 \text{ TeV}$. This is below both $1/2$ and $1/6$.

The high-energy contact limit in the same four-state basis has maximum eigenvalue magnitude 0.02044286, so the endpoint behavior approaches the independently generated quartic asymptote as expected.

MODEL-DEPENDENT RESULT

The chosen benchmark has no physical-scalar perturbative-unitarity failure below 20 TeV. This does not prove that the EFT cutoff exceeds 20 TeV, because unspecified higher-dimensional operators may enter earlier.

11.21 6. Goldstone-equivalence scan

Following the established scalar-sector method,^{1,2} the four zero modes were assigned Feynman-gauge masses:

- two charged SM Goldstones: 80.4 GeV;
- one neutral SM Goldstone: 91.2 GeV;
- one dark Goldstone: $m_X = 3g_X f = 600 \text{ GeV}$ for $g_X = 0.20$.

All 36 unordered scalar/Goldstone channels were included. The scan began above the heaviest pair threshold and extended to 20 TeV. After partial removal of channels with angular poles, the maximum was

$$\max_{\sqrt{s}} \max_i |a_0^i| = 0.02396323$$

at 20 TeV. All 36 channels were retained at that energy. The value approaches the complete eight-real-scalar high-energy result 0.02454469.

This calculation neglects gauge vertices. It is therefore a gaugeless Goldstone-equivalence test, not a full longitudinal- and transverse-vector scattering calculation.

11.22 7. Cutoff interpretation

The present result establishes only that the tested renormalizable scalar interactions do not force perturbative breakdown within the scanned range. It does not determine the Wilsonian cutoff.

For the minimal hidden scalars,

$$(16\pi^2)\beta_{g_X} = \frac{10}{3}g_X^3.$$

The gauge-only one-loop Landau-pole estimate is

$$\Lambda_{LP} = \mu_0 \exp\left[\frac{8\pi^2}{(10/3)g_X^2(\mu_0)}\right].$$

For $g_X = 0.20$, this formal scale is exponentially remote. It is not physically meaningful to extrapolate it without the complete coupled RG system, threshold matching, gravitational effects, and a UV completion.

11.23 8. Referee verdict

Question	Status
Correct identical-particle normalization	Verified against published analytic eigenvalues
Contact plus all scalar exchanges	Completed
Open-channel thresholds	Completed
Explicit s -channel exclusion	Completed
Explicit t/u -pole detection	Completed
Physical-scalar scan	Passed for benchmark
Gaugeless scalar/Goldstone scan	Passed for benchmark
Gauge-parameter independence	Not shown
Full vector/scalar coupled matrix	Open
Independent package reproduction	Open
Universal TP-1 parameter-space bound	Not shown
EFT cutoff determination	Open

The finite-energy gate is therefore **passed provisionally for the selected benchmark**, not closed for the theory as a whole.

11.24 Reproducibility

- Python 3
- NumPy 2.3.5
- SciPy 1.17.0
- Gauss–Legendre orders: 48, 96, 192
- Main scan order: 96
- s -channel cut: $C_S = 0.25$
- Physical-scalar scan: 273 evaluated energies
- Goldstone scan: 220 energies
- Upper scan energy: 20 TeV

The implementation is contained in `finite_energy_unitarity.py`. The independent published-limit check is `validate_finite_energy_against_hektor.py`.

11.25 Sources

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2. A. Hektor, A. Hryczuk, and K. Kannike, “[Improved bounds on \$Z_3\$ singlet dark matter](#),” *Journal of High Energy Physics* 2019, 204 (2019).
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11.26 S3 Ternary Polarity Phase II — Kernel-Distinctness Audit

Project: *Ternary Polarity: Toward a Z_3 Framework for Fundamental Physics*

Author: Sushabhan Giri

Audit date: 15 September 2026

Canonical branch tested: TP-D

Scientific status: speculative field-theory construction under audit; not an established theory of nature

11.27 Executive verdict

SUBSTANTIVE PARTIAL PASS

The dual charge assignment

$$A \sim (1, 0), \quad B \sim (0, 1)$$

under $G_{TP} = Z_3^{(m)} \times Z_3^{(E)}$ is physically more restrictive than assigning both fields charge one under only the projected group $Z_3^{(P)}$. The extra restriction is not erased by a change of field basis: it is equivalently the existence of a noncentral order-three symmetry generator whose charge is conserved by the full action. It forbids a definite set of kinetic, mass, cubic, quartic, and Higgs-portal operators.

The strongest all-orders null prediction of the minimal exact model is

$$\mathcal{M}(B \rightarrow A + X_0) = 0$$

for any final collection X_0 neutral under the kernel symmetry, provided A and B are nondegenerate kernel-charge eigenstates and the full dual symmetry is exact. In particular,

$$\Gamma(B \rightarrow A + h) = 0.$$

An ordinary single- Z_3 theory permits the off-diagonal portal $(H^\dagger H)A^\dagger B + \text{h.c.}$, so it generically permits this decay when kinematically open.

This passes the mathematical nonredundancy and radiative-protection tests. It does **not** yet pass the strong novelty or phenomenological-identifiability tests. Product discrete symmetries, symmetry-protected operator zeros, multi-component dark sectors, and semi-annihilation are established model-building tools. A generic single- Z_3 comparator can also tune its allowed off-diagonal couplings to zero. Consequently a null decay alone cannot positively identify TP-D, although observing a nonzero kernel-changing decay would falsify the exact minimal realization.

The branch therefore advances to the next research stage only as a **kernel-sensitive** $Z_3 \times Z_3$ **EFT**, not as a demonstrated new fundamental theory.

11.28 Document A — Kernel and symmetry analysis

Define

$$\pi : Z_3^{(m)} \times Z_3^{(E)} \longrightarrow Z_3^{(P)}, \quad \pi(q_m, q_E) = q_m + q_E \pmod{3}.$$

Then

$$\ker \pi = \{(0, 0), (1, 2), (2, 1)\} \cong Z_3.$$

The map is surjective and gives the split exact sequence

$$0 \longrightarrow Z_3 \longrightarrow Z_3^2 \xrightarrow{\pi} Z_3 \longrightarrow 0.$$

A useful invertible change of charge coordinates over $\mathbb{F}_3$ is

$$p = q_m + q_E, \quad k = q_m - q_E,$$

because

$$\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2 \equiv 1 \pmod{3}.$$

Thus the dual group can be written as $Z_3^{(P)} \times Z_3^{(K)}$. In this basis,

$$A : (p, k) = (1, 1), \quad B : (p, k) = (1, 2).$$

This establishes two facts that must be kept separate:

1. **Exact result:** A and B have the same projected charge but different kernel charge.
2. **Novelty limitation:** the abstract group is the standard direct product $Z_3 \times Z_3$. The names “matter-like” and “energy-like” do not add physics unless an independent construction gives those labels operational meaning.

Strictly, charges label characters while group elements act on fields. Because finite Abelian groups are isomorphic to their character groups, one may choose a self-dual pairing, but the identification is a convention. The physical statement is the existence of the extra transformation

$$K = \text{diag}(\omega, \omega^2), \quad \omega = e^{2\pi i/3},$$

acting on $\Phi = (A, B)^T$. The projected generator is $P = \omega I_2$.

11.28.1 Kernel pass/fail result

If the action is invariant only under P , the kernel is physically unobserved. If it is invariant under both P and K , amplitudes must conserve both charges. The minimal TP-D action below is invariant under K ; therefore the kernel acts nontrivially.

11.29 Documents B and C — Exact TP-D and generic- Z_3 Lagrangians

Assume that A and B are Lorentz scalars and Standard-Model gauge singlets, while all Standard-Model fields are neutral under the new discrete factors. Natural units $\hbar = c = 1$ are used. Scalar fields have mass dimension one.

11.29.1 Generic projected- Z_3 comparator

Let $\Phi = (A, B)^T$, with both components transforming as $\Phi \rightarrow \omega \Phi$. The most general renormalizable scalar Lagrangian is

$$\mathcal{L}_{Z_3}^{\text{generic}} = (\partial_\mu \Phi)^\dagger Z (\partial^\mu \Phi) - V_{Z_3},$$

where Z is a positive Hermitian 2×2 kinetic matrix. It can be made the identity by an invertible field redefinition. In a canonical kinetic basis,

$$\begin{aligned}
V_{Z_3} = & V_H + m_A^2 |A|^2 + m_B^2 |B|^2 + (m_{AB}^2 A^\dagger B + \text{h.c.}) \\
& + \left[\frac{\mu_A}{3} A^3 + \mu_{AAB} A^2 B + \mu_{ABB} A B^2 + \frac{\mu_B}{3} B^3 + \text{h.c.} \right] \\
& + \lambda_A |A|^4 + \lambda_B |B|^4 + \lambda_{AB} |A|^2 |B|^2 \\
& + \left[\eta_A |A|^2 A^\dagger B + \eta_B |B|^2 A^\dagger B + \frac{\eta_2}{2} (A^\dagger B)^2 + \text{h.c.} \right] \\
& + (H^\dagger H) [\lambda_{HA} |A|^2 + \lambda_{HB} |B|^2 + (\lambda_{HAB} A^\dagger B + \text{h.c.})].
\end{aligned}$$

Here $V_H = -m_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2$. Hermiticity makes $m_A^2, m_B^2, \lambda_A, \lambda_B, \lambda_{AB}, \lambda_{HA}, \lambda_{HB}$ real; the displayed phase-sensitive coefficients may be complex.

11.29.2 Exact dual TP-D

Requiring the full $Z_3^{(m)} \times Z_3^{(E)}$ gives

$$\begin{aligned}
\mathcal{L}_{\text{TP-D}} &= |\partial A|^2 + |\partial B|^2 - V_{\text{TP-D}}, \\
V_{\text{TP-D}} &= V_H + m_A^2 |A|^2 + m_B^2 |B|^2 + \left(\frac{\mu_A}{3} A^3 + \frac{\mu_B}{3} B^3 + \text{h.c.} \right) \\
&+ \lambda_A |A|^4 + \lambda_B |B|^4 + \lambda_{AB} |A|^2 |B|^2 \\
&+ (H^\dagger H) (\lambda_{HA} |A|^2 + \lambda_{HB} |B|^2).
\end{aligned}$$

Nonzero μ_A and μ_B are important: without them, the renormalizable potential has the larger accidental symmetry $U(1)_A \times U(1)_B$. The two cubics reduce this to the intended $Z_3 \times Z_3$.

11.30 Document D — Complete distinguishing operator table

The enumerator treats Hermitian-conjugate pairs as one operator class. It finds 3 projected- Z_3 -invariant quadratic, 4 cubic, and 6 quartic nonderivative classes. The six comparator-only nonderivative classes are exactly those shown below.

Operator class	Dimension	Generic $Z_3^{(P)}$	Exact TP-D	Consequence if present
$\partial A^\dagger \partial B + \text{h.c.}$	4	allowed	forbidden	kinetic flavour mixing
$A^\dagger B + \text{h.c.}$	2	allowed	forbidden	mass mixing and oscillation
$A^2 B + \text{h.c.}$	3	allowed	forbidden	conversion/decay vertices
$A B^2 + \text{h.c.}$	3	allowed	forbidden	conversion/decay vertices
$ A ^2 A^\dagger B + \text{h.c.}$	4	allowed	forbidden	flavour-changing scattering
$ B ^2 A^\dagger B + \text{h.c.}$	4	allowed	forbidden	flavour-changing scattering
$(A^\dagger B)^2 + \text{h.c.}$	4	allowed	forbidden	pair conversion
$(H^\dagger H) A^\dagger B + \text{h.c.}$	4	allowed	forbidden	Higgs-mediated transition/decay

The common nonderivative classes are

$$|A|^2, |B|^2, A^3 + \text{h.c.}, B^3 + \text{h.c.}, |A|^4, |B|^4, |A|^2|B|^2.$$

The Standard-Model Higgs adds the common portals $(H^\dagger H)|A|^2$ and $(H^\dagger H)|B|^2$. No renormalizable linear singlet portal is allowed because A and B have nonzero projected charge.

The enumeration is exhaustive for nonderivative monomials in two commuting complex singlets through dimension four. Gauge-field and Standard-Model-only operators are common to both theories and cancel from the comparator.

Therefore

$$\Delta\mathcal{L} = \mathcal{L}_{Z_3}^{\text{generic}} - \mathcal{L}_{\text{TP-D}}$$

is precisely the set of cross terms in the table, modulo field-basis normalization.

11.31 Document E — Spectrum and mixing analysis

In the generic comparator, the quadratic matrix is

$$M^2 = \begin{pmatrix} m_A^2 & m_{AB}^2 \\ m_{AB}^{2*} & m_B^2 \end{pmatrix}.$$

After one relative rephasing makes m_{AB}^2 real and nonnegative,

$$m_\pm^2 = \frac{m_A^2 + m_B^2}{2} \pm \frac{1}{2} \sqrt{(m_B^2 - m_A^2)^2 + 4|m_{AB}^2|^2},$$

and

$$\tan 2\theta = \frac{2|m_{AB}^2|}{m_B^2 - m_A^2}.$$

For $|m_{AB}^2| \ll |m_B^2 - m_A^2|$,

$$\theta \simeq \frac{|m_{AB}^2|}{m_B^2 - m_A^2}.$$

Exact TP-D enforces

$$m_{AB}^2 = 0, \quad \theta = 0.$$

The numerical diagonalization test agreed with the analytic eigenvalues to $1.2 \times 10^{-10} \text{ GeV}^2$ for the stored test point; 100 randomized regression points passed.

11.31.1 Oscillation statement

For coherent relativistic propagation in a broken model,

$$P(A \rightarrow B; L) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 L}{4E}\right).$$

In exact TP-D, $P(A \rightarrow B; L) = 0$. This is meaningful only if production and detection define the interaction basis and coherence is maintained. It must not be advertised as a generic prediction before an experimental realization is specified.

11.32 Basis-invariance audit

The statement “the off-diagonal entry vanishes” is basis dependent. The physically invariant statement is that there exists a noncentral unitary matrix K , with

$$K^3 = I, \quad K \not\propto I,$$

that leaves **every** coupling tensor invariant. For the quadratic tensor,

$$K^\dagger M^2 K = M^2.$$

Under a basis transformation $\Phi \rightarrow U\Phi$,

$$M^2 \rightarrow U M^2 U^\dagger, \quad K \rightarrow U K U^\dagger,$$

so the existence of the symmetry is unchanged. One diagnostic is

$$I_M(K) = \text{Tr}[(K^\dagger M^2 K - M^2)^\dagger (K^\dagger M^2 K - M^2)].$$

Exact invariance requires $I_M = 0$, together with analogous zero residuals for the cubic, quartic, and portal tensors.

A mass matrix by itself is insufficient: one can diagonalize any Hermitian matrix and invent a diagonal order-three matrix in that basis. The nontrivial test is whether the **same** K leaves all interaction tensors invariant. TP-D passes this simultaneous-tensor test by construction. The generic comparator does not pass it for generic coefficients.

The matrix element between nondegenerate charge eigenstates,

$$\langle A; X_0 | S | B \rangle,$$

is an observable. If $[S, K] = 0$ and X_0 is kernel neutral, its initial and final kernel phases differ, forcing the amplitude to zero. That is the basis-independent content of the selection rule.

11.33 Parameter-count audit

Shared Standard-Model parameters are omitted.

11.33.1 Generic projected Z_3

After canonicalizing the kinetic term:

- Hermitian mass matrix: 4 real parameters;
- complex symmetric cubic rank-three tensor in two flavours: 8;
- Hermitian quartic tensor on $\text{Sym}^2(\mathbb{C}^2)$: 9;
- Hermitian Higgs-portal matrix: 4.

This gives 25 real coefficients. A common $U(2)$ field-basis transformation removes four redundancies, leaving

$$N_{phys}^{Z_3} = 21$$

for a generic point with no accidental stabilizer.

11.33.2 Exact TP-D

The exact dual potential has 2 real masses, 2 complex cubics, 3 real dark quartics, and 2 real diagonal Higgs portals: 11 real coefficients. Two independent field rephasings remove the two cubic phases, leaving

$$N_{phys}^{TP-D} = 9.$$

Thus this realization removes twelve physical continuous parameters relative to the most general projected comparator:

$$\Delta N_{phys} = 12.$$

This is real parameter compression, but most of it consists of symmetry-protected zeros. No nonzero numerical sum rule has yet been derived.

11.34 Document F — Vacuum and stability audit

The exact dark quartic is

$$V_4 = \lambda_A |A|^4 + \lambda_B |B|^4 + \lambda_{AB} |A|^2 |B|^2.$$

Its exact tree-level bounded-from-below conditions are

$$\lambda_A > 0, \quad \lambda_B > 0, \quad \lambda_{AB} > -2\sqrt{\lambda_A \lambda_B},$$

with non-strict boundary variants requiring separate flat-direction analysis.

For one field, after minimizing its phase,

$$V_i(r) = m_i^2 r^2 - \frac{2|\mu_i|}{3} r^3 + \lambda_i r^4.$$

The origin is no higher than every nonzero stationary point when

$$|\mu_i|^2 \leq 9\lambda_i m_i^2.$$

Therefore the following are simple sufficient conditions for the two-field origin to be the global minimum:

$$\lambda_{AB} \geq 0, \quad |\mu_A|^2 \leq 9\lambda_A m_A^2, \quad |\mu_B|^2 \leq 9\lambda_B m_B^2,$$

in addition to positive masses and self-quartics. They are sufficient, not necessary. A negative portal requires a full mixed stationary-point analysis.

The stored benchmark

$$(m_A, m_B, |\mu_A|, |\mu_B|) = (180, 400, 60, 90) \text{ GeV},$$

$$(\lambda_A, \lambda_B, \lambda_{AB}) = (0.20, 0.25, 0.10)$$

passes these sufficient conditions. A deterministic 200,000-point log-radial numerical search with seed 20260915 found no negative value after phase minimization. This numerical scan corroborates the analytic sufficient proof; it is not the proof itself.

Higgs-portal terms shift the physical masses after electroweak symmetry breaking:

$$M_A^2 = m_A^2 + \frac{1}{2}\lambda_{HA}v^2, \quad M_B^2 = m_B^2 + \frac{1}{2}\lambda_{HB}v^2.$$

The full electroweak-plus-dark global-vacuum audit remains to be performed for a final phenomenology benchmark.

11.35 Document G — High-energy unitarity audit

Writing

$$A = (a_1 + ia_2)/\sqrt{2}, \quad B = (b_1 + ib_2)/\sqrt{2},$$

the normalized ten-channel real-scalar quartic scattering matrix has eigenvalues

$$2\lambda_A \quad (\times 2), \quad 2\lambda_B \quad (\times 2), \quad \lambda_{AB} \quad (\times 4),$$

and

$$2(\lambda_A + \lambda_B) \pm \sqrt{4(\lambda_A - \lambda_B)^2 + \lambda_{AB}^2}.$$

With the convention $a_0 = -\Lambda/(16\pi)$, tree-level perturbative unitarity requires every eigenvalue Λ to satisfy

$$|\Lambda| \leq 8\pi.$$

For the stored benchmark the spectral radius is 1.041421356, far below 8π . The analytic spectrum was verified against the directly constructed 10×10 tensor matrix at 100 randomized coupling points with relative and absolute tolerances 10^{-12} .

Scope limitation: this is the high-energy dark-scalar quartic limit. A final benchmark still requires Higgs/Goldstone channels and finite-energy exchange diagrams with pole excision. Thus the complete finite-energy unitarity item is open.

11.36 Document H — Radiative-stability audit

Every exact-TP-D vertex has total vector charge (0,0). Charge adds at internal contractions, so every perturbative diagram with external legs of nonzero total vector charge vanishes. Equivalently, a symmetry-preserving regulator and subtraction scheme produce a renormalized effective action satisfying

$$\Gamma[\omega A, B] = \Gamma[A, B], \quad \Gamma[A, \omega B] = \Gamma[A, B].$$

Consequently no counterterm proportional to

$$A^\dagger B, \quad A^2 B, \quad AB^2, \quad |A|^2 A^\dagger B, \quad |B|^2 A^\dagger B, \quad (A^\dagger B)^2, \quad (H^\dagger H) A^\dagger B$$

can be generated while the full symmetry is exact. In particular,

$$\beta_{m_{AB}^2} \Big|_{\text{all kernel-breaking spurions}=0} = 0.$$

This conclusion holds to all perturbative orders. The scalar-only model has no chiral gauge anomaly. A parent gauge completion containing chiral fermions would require a separate continuous and residual-discrete anomaly audit.

11.36.1 Controlled breaking TP-B

Introduce a scalar spurion/field

$$\Sigma \sim (1, 2), \quad q_P(\Sigma) = 0.$$

A vacuum expectation value $\langle \Sigma \rangle = w/\sqrt{2}$ preserves the diagonal projected subgroup but breaks the kernel symmetry. The renormalizable term

$$\mu_\Sigma \Sigma A^\dagger B + \text{h.c.}$$

generates

$$m_{AB}^2 = \frac{\mu_\Sigma w}{\sqrt{2}}.$$

Similarly, $\Sigma A^2 B$ and $\Sigma^\dagger A B^2$ generate mixed cubics linearly in w . Off-diagonal kinetic, quartic, and Higgs-portal terms arise through dimension-five operators and scale as w/Λ in the minimal EFT. Hence

$$\theta \sim \frac{\mu_\Sigma w}{\sqrt{2}(m_B^2 - m_A^2)}, \quad \Gamma_{K\text{-violating}} \propto w^2$$

at leading order, provided no lower-order breaking source exists. The coefficients remain independent Wilson coefficients; symmetry alone does not give a numerical sum rule among them.

11.37 Document I — Minimal falsification benchmark

Take the illustrative masses

$$M_A = 180 \text{ GeV}, \quad M_B = 400 \text{ GeV}, \quad m_h = 125.25 \text{ GeV},$$

so $B \rightarrow A + h$ is kinematically open. In the generic comparator, after electroweak symmetry breaking,

$$(H^\dagger H) A^\dagger B + \text{h.c.} \supset \lambda_{HAB} v h A^\dagger B + \text{h.c.}$$

and

$$\Gamma(B \rightarrow Ah) = \frac{|\lambda_{HAB} v|^2}{16\pi M_B} \lambda^{1/2} \left(1, \frac{M_A^2}{M_B^2}, \frac{m_h^2}{M_B^2} \right).$$

For the purely illustrative comparator value $\lambda_{HAB} = 0.01$, the stored calculation gives

$$\Gamma_{Z_3}(B \rightarrow Ah) = 1.93024 \times 10^{-4} \text{ GeV},$$

whereas exact TP-D gives zero.

This number is not a TP prediction; $\lambda_{HAB} = 0.01$ was chosen only to display an effect size. The robust prediction is the exact zero.

11.37.1 Falsification card

Under the assumptions that:

1. the observed states are the two nondegenerate TP-D charge eigenstates;
2. the visible final state is kernel neutral;
3. no explicit or spontaneous kernel breaking occurs;
4. the tested decay is kinematically allowed;

then

Any statistically significant nonzero $B \rightarrow A + h$ amplitude falsifies exact minimal TP-D.

A null result does not prove TP-D because the generic comparator may set $\lambda_{HAB} = 0$ accidentally. Positive model identification would require observing both species and enough independent allowed and forbidden channels to infer the extra conserved charge.

11.38 Document J — Preliminary novelty matrix

Feature	Generic single Z_3	Established enlarged discrete models	TP-D result	Novelty status
residual charge-one fields	yes	yes	yes	known
cubic Z_3 self-interactions	yes	yes	A^3, B^3	known
semi-annihilation	yes	yes	allowed within each sector	known
multiple stable species	possible	established	natural if each is lightest in its vector-charge sector	known mechanism
extra conserved kernel charge	no	yes for product groups	yes	known group mechanism
forbidden $A^\dagger B$	not generic	common when charges differ under extra factor	yes	known selection-rule consequence
all-orders null $B \rightarrow A + X_0$	not generic	standard consequence of an exact extra charge	yes	known mechanism, TP-specific labeling
$q_P = q_m + q_E$ projection	optional relabeling	homomorphisms are standard	central organizing map	known mathematics; combination-specific
nonzero numerical sum rule	none required	model dependent	none found	absent
unique experimental signature	no	model dependent	not yet demonstrated	open

Relevant prior work already establishes the nearby physics:

- G. Bélanger, K. Kannike, A. Pukhov, and M. Raidal, “[Impact of semi-annihilations on dark matter phenomenology—an example of \$Z_N\$ -symmetric scalar dark matter](#)”, develops discrete-charge operator selection, semi-annihilation, and enlarged-symmetry multicomponent sectors.
- P. Ko and Y. Tang, “[Self-interacting scalar dark matter with local \$Z_3\$ symmetry](#)”, realizes local Z_3 as a remnant of a broken dark $U(1)$.
- C. E. Yaguna and Ó. Zapata, “[Multi-component scalar dark matter from a \$Z_N\$ symmetry: a systematic analysis](#)”, shows that multiple stable scalar species and nontrivial conversion patterns already arise in established discrete-symmetry models.
- D. Borah, E. Ma, and D. Nanda, “[Dark \$SU\(2\) \rightarrow Z_3 \times Z_2\$ Gauge Symmetry](#)”, is a concrete example of product residual discrete symmetry used to organize multiple dark states.

Therefore the present construction has **not** passed a claim of demonstrable literature novelty. Its defensible contribution at this checkpoint is a precise TP-specific formulation and comparator audit, not discovery of product-discrete dark matter.

11.39 Document K — Gate report

11.39.1 1. Executive verdict

SUBSTANTIVE PARTIAL PASS

11.39.2 2. What survived

- The projection has a nontrivial three-element kernel.
- The minimal two-field representation makes the kernel act nontrivially.
- The complete renormalizable operator basis differs from the generic projected- Z_3 basis.
- The distinction is expressible as a basis-invariant symmetry of all coupling tensors.
- Kernel-changing amplitudes vanish to all perturbative orders in the exact model.
- A healthy inert tree-level benchmark exists under explicit sufficient conditions.
- The dark-scalar high-energy unitarity matrix is perturbative at that benchmark.
- Exact TP-D removes twelve physical continuous parameters relative to the generic comparator under the stated counting assumptions.

11.39.3 3. What failed or remains incomplete

- No unique nonzero numerical relation or sum rule has been found.
- A null transition is not positive evidence because generic Z_3 can tune the corresponding coupling to zero.
- The core group and its selection-rule mechanism are already known mathematics and model-building technology.
- Full electroweak-plus-dark vacuum analysis is unfinished.
- Full finite-energy scalar/Higgs/Goldstone/gauge unitarity is unfinished.
- The complete one-loop RGE system for TP-D has not yet been independently regenerated, although the forbidden-operator surface is protected exactly by symmetry.
- No relic-density, collider-production, detector-background, or global-likelihood analysis has yet established practical accessibility.
- A gauge-origin and anomaly-free completion has not yet been constructed.

11.39.4 4. What is genuinely new

No element is yet demonstrably new after literature comparison. The potentially original contribution is the **specific interpretation and systematic use of the projection/kernel pair as the defining TP architecture**, together with a future overconstrained observable pattern. That claim remains “known ingredients, newly combined” until a broader search and a distinctive prediction succeed.

11.39.5 5. What is known physics

Finite Abelian product groups, residual discrete symmetries, multicomponent stable sectors, cubic Z_3 interactions, semi-annihilation, symmetry-protected coupling zeros, and forbidden-transition null tests are known.

11.39.6 6. First distinctive prediction

For exact minimal TP-D,

$$\mathcal{M}(B \rightarrow A + X_0) = 0$$

whenever X_0 is kernel neutral. This differs from the generic single- Z_3 theory, which allows the amplitude.

11.39.7 7. Experimental falsification condition

Observation of any kernel-changing transition, such as $B \rightarrow A + h$, excludes exact TP-D under the explicit four assumptions in the falsification card.

11.39.8 8. Remaining blockers, ranked

1. **Critical:** find an identifiable pattern or nonzero relation that generic Z_3 cannot imitate by tuning.
2. **Critical:** establish production and detection channels for both charge eigenstates.
3. **High:** complete current-data phenomenology and coupled relic-density analysis.
4. **High:** finish electroweak vacuum and finite-energy unitarity audits.
5. **High:** independently derive the full TP-D one-loop RGEs.
6. **Medium:** construct and audit a minimal residual-gauge completion.
7. **Medium:** complete a systematic prior-art search for the exact operator basis and signature.

11.39.9 9. Confidence assessment

Domain	Confidence	Reason
finite-group and operator algebra	high	exhaustive enumeration plus analytic charge proof
mass and basis analysis	high	exact formulas and randomized numerical cross-checks
tree-level dark vacuum	medium-high	exact BFB and sufficient global-origin proof; not full Higgs system
high-energy dark-scalar unitarity	high within stated scope	direct tensor construction and randomized spectral test
all-orders selection-rule protection	high	exact finite symmetry and scalar-only anomaly status
phenomenology	low-medium	one illustrative decay; no production/global constraint study
novelty	low	nearby mechanisms are established; exact combination not exhaustively searched
experimental accessibility	low-medium	possible in principle, no detector-level analysis

11.39.10 10. Recommendation

Continue TP-D conditionally; do not promote it beyond a $Z_3 \times Z_3$ EFT.

The immediate next objective is not gravity or grand unification. It is to construct a minimal visible/dark portal with more measured quantities than free parameters and search for a correlated set of kernel-conserving and kernel-violating rates. If no such overconstrained pattern exists, TP-D remains a valid conventional product-discrete EFT rather than a new theoretical framework.

11.40 Reproducibility record

- Operator enumerator: `code/dual_tp_operator_enumerator.py`
- Machine-readable operator basis: `audits/dual_tp_operator_basis.json`
- Vacuum, spectrum, unitarity, and decay validator: `code/dual_tp_validation.py`
- Machine-readable numerical output: `audits/dual_tp_validation.json`
- Regression tests: `code/test_dual_tp.py`
- Test result: 5/5 passed on 15 September 2026.

The numerical vacuum scan used NumPy 2.3.5, 200,000 samples, seed 20260915, and log-uniform radial sampling. The analytic inequalities, not the scan, establish the stated sufficient vacuum result.

11.41 S4 Ternary Polarity checkpoint 6

11.42 TP-D electroweak vacuum, scalar unitarity, and one-loop running

Research status: internal technical audit

Theory version: TP-D v0.1

Date: 15 September 2026

Epistemic level: exact tree-level results plus model-dependent one-loop results

11.43 Executive result

The minimal kernel-sensitive model passes three limited but important tests:

1. its benchmark electroweak vacuum is the global tree-level minimum under an explicit analytic sufficient condition;
2. its complete high-energy scalar/Goldstone quartic scattering matrix satisfies tree-level partial-wave unitarity at the benchmark;
3. its 15-channel finite-energy physical-scalar matrix satisfies tree-level partial-wave unitarity from threshold to 20 TeV after including every scalar contact and exchange diagram;
4. its scalar-only one-loop quartic beta functions agree coefficient by coefficient in two independent derivations.

The first one-loop running test also produced a negative result. The weak-portal benchmark **B0** loses tree-level boundedness along the running couplings at approximately

$$\mu_{BFB} = 3.57 \times 10^8 \text{ GeV}.$$

A diagnostic stronger-portal benchmark **B1** avoids that failure over the perturbative interval, but reaches the scalar-unitarity boundary near

$$\mu_U = 7.18 \times 10^{14} \text{ GeV}$$

and the largest-coupling 4π boundary near

$$\mu_{4\pi} = 1.35 \times 10^{15} \text{ GeV}.$$

Therefore this checkpoint is **not** a proof of ultraviolet completion or phenomenological viability. It establishes a healthy tree-level example and a controlled one-loop interval, while exposing a high-scale trade-off between Higgs-quartic stabilization and perturbativity.

11.44 1. Model and conventions

The exact dual symmetry is

$$G_{TP-D} = Z_3^{(m)} \times Z_3^{(E)}, \quad A \sim (1, 0), \quad B \sim (0, 1),$$

with Standard Model fields neutral. In the unbroken inert phase, the most general renormalizable scalar potential is

$$\begin{aligned} V = & -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2 + m_A^2 |A|^2 + m_B^2 |B|^2 \\ & + \left(\frac{\mu_A}{3} A^3 + \frac{\mu_B}{3} B^3 + \text{h.c.} \right) + \lambda_A |A|^4 + \lambda_B |B|^4 \\ & + \lambda_{HA} (H^\dagger H) |A|^2 + \lambda_{HB} (H^\dagger H) |B|^2 + \lambda_{AB} |A|^2 |B|^2. \end{aligned}$$

The phases of A and B can independently minimize their cubic terms. With

$$H^\dagger H = \frac{h^2}{2}, \quad |A|^2 = \frac{a^2}{2}, \quad |B|^2 = \frac{b^2}{2},$$

and nonnegative a, b, h , the phase-minimized potential used in the audit is

$$V_{\min} = -\frac{\mu_H^2 h^2}{2} + \frac{\lambda_H h^4}{4} + \frac{m_A^2 a^2}{2} - \frac{|\mu_A| a^3}{3\sqrt{2}} + \frac{\lambda_A a^4}{4} \\ + \frac{m_B^2 b^2}{2} - \frac{|\mu_B| b^3}{3\sqrt{2}} + \frac{\lambda_B b^4}{4} + \frac{\lambda_{HA} h^2 a^2 + \lambda_{HB} h^2 b^2 + \lambda_{AB} a^2 b^2}{4}.$$

All formulas use natural units, $\hbar = c = 1$. The scalar fields and cubic coefficients have mass dimension one; squared-mass parameters have dimension two; quartics are dimensionless.

11.45 2. Exact bounded-from-below test

Let

$$x = H^\dagger H, \quad y = |A|^2, \quad z = |B|^2,$$

so that

$$V_4 = \lambda_H x^2 + \lambda_A y^2 + \lambda_B z^2 + \lambda_{HA} xy + \lambda_{HB} xz + \lambda_{AB} yz.$$

This is a quadratic form on the nonnegative orthant with symmetric matrix

$$Q = \begin{pmatrix} \lambda_H & \lambda_{HA}/2 & \lambda_{HB}/2 \\ \lambda_{HA}/2 & \lambda_A & \lambda_{AB}/2 \\ \lambda_{HB}/2 & \lambda_{AB}/2 & \lambda_B \end{pmatrix}.$$

For a real symmetric 3×3 matrix, the exact copositivity conditions can be written as

$$\lambda_H, \lambda_A, \lambda_B \geq 0,$$

$$\bar{q}_{ij} = q_{ij} + \sqrt{q_{ii} q_{jj}} \geq 0,$$

and

$$\sqrt{q_{11} q_{22} q_{33}} + q_{12} \sqrt{q_{33}} + q_{13} \sqrt{q_{22}} + q_{23} \sqrt{q_{11}} + \sqrt{2 \bar{q}_{12} \bar{q}_{13} \bar{q}_{23}} \geq 0.$$

The implementation follows the copositivity formulation reviewed by [Kannike](#). For B0 the smallest pairwise margin is 0.17086235, the final three-field margin is 0.24509872, and all exact conditions pass.

11.46 3. Analytic proof of the benchmark global minimum

The benchmark uses

$$v = 246.22 \text{ GeV}, \quad m_h = 125.25 \text{ GeV}, \quad M_A = 180 \text{ GeV}, \quad M_B = 400 \text{ GeV},$$

$$\lambda_A = 0.20, \quad \lambda_B = 0.25, \quad \lambda_{AB} = 0.10, \quad \lambda_{HA} = 0.020, \quad \lambda_{HB} = 0.030,$$

$$|\mu_A| = 60 \text{ GeV}, \quad |\mu_B| = 90 \text{ GeV}.$$

The Lagrangian masses are fixed by

$$m_A^2 = M_A^2 - \frac{\lambda_{HA}v^2}{2}, \quad m_B^2 = M_B^2 - \frac{\lambda_{HB}v^2}{2}.$$

For either dark radial field r ,

$$U(r) = \frac{m^2 r^2}{2} - \frac{|\mu| r^3}{3\sqrt{2}} + \frac{\lambda r^4}{4} = r^2 \left(\frac{m^2}{2} - \frac{|\mu| r}{3\sqrt{2}} + \frac{\lambda r^2}{4} \right).$$

The expression in parentheses is nonnegative for every $r \geq 0$ if

$$|\mu|^2 \leq 9\lambda m^2.$$

Because all three portal quartics are nonnegative at B0, and both dark inequalities hold, every dark-field contribution relative to $a = b = 0$ is nonnegative. It follows directly that

$$V_{\min}(h, a, b) \geq -\frac{\mu_H^2 h^2}{2} + \frac{\lambda_H h^4}{4},$$

whose global minimum is $(h, a, b) = (v, 0, 0)$. This is an analytic sufficient proof, not only a numerical scan.

At this point the radial Hessian eigenvalues are

$$(m_h^2, M_A^2, M_B^2) = (15687.5625, 32400, 160000) \text{ GeV}^2.$$

A 300-start stationary-point search found only the electroweak minimum and the unstable origin within the stated search region. Differential evolution reproduced the electroweak vacuum energy to $5.8 \times 10^{-6} \text{ GeV}^4$. These numerical results check, but do not replace, the analytic proof.

11.47 4. Complete high-energy scalar/Goldstone unitarity matrix

Writing the Higgs doublet as four real fields and each complex singlet as two real fields gives eight real scalar components and 36 normalized two-particle channels. The code constructs the full quartic tensor and diagonalizes the corresponding 36×36 scattering matrix.

Its spectrum reduces exactly to

- $2\lambda_H$, multiplicity 9;
- $2\lambda_A$, multiplicity 2;
- $2\lambda_B$, multiplicity 2;
- λ_{HA} , multiplicity 8;
- λ_{HB} , multiplicity 8;
- λ_{AB} , multiplicity 4;
- three eigenvalues of

$$S = \begin{pmatrix} 6\lambda_H & \sqrt{2}\lambda_{HA} & \sqrt{2}\lambda_{HB} \\ \sqrt{2}\lambda_{HA} & 4\lambda_A & \lambda_{AB} \\ \sqrt{2}\lambda_{HB} & \lambda_{AB} & 4\lambda_B \end{pmatrix}.$$

With $a_0 = -\Lambda/(16\pi)$, the tree criterion $|\text{Re } a_0| \leq 1/2$ becomes $|\Lambda| \leq 8\pi$. B0 has

$$\max |\Lambda| = 1.05056, \quad \max |a_0| = 0.02090,$$

and passes comfortably. A 250-point random comparison between direct tensor diagonalization and the analytic spectrum found a maximum absolute eigenvalue discrepancy of 8.0×10^{-15} .

This result is restricted to the high-energy quartic limit. Finite-energy scalar exchange is added below; a gauge-complete analysis must additionally establish Goldstone/longitudinal-vector consistency, include transverse gauge states where relevant, and declare its pole treatment. The methodological need for finite-energy testing is discussed by [Goodsell and Staub](#).

11.47.1 4.1 Finite-energy physical-scalar result

The physical external states at B0 are

$$h, A_1, A_2, B_1, B_2$$

with masses (125.25, 180, 180, 400, 400) GeV. Their 15 unordered two-particle channels were scanned from the lightest threshold, 250.5025 GeV, to 20 TeV. The matrix includes the quartic contact term and every scalar s -, t -, and u -channel exchange:

$$\mathcal{M}_{ab \rightarrow cd} = -\lambda_{abcd} - \sum_e \frac{g_{abe}g_{cde}}{s - m_e^2} - \sum_e \frac{g_{ace}g_{bde}}{t - m_e^2} - \sum_e \frac{g_{ade}g_{bce}}{u - m_e^2}.$$

The scan used the same partial-wave normalization as the high-energy audit. No sampled point required an s -channel or angular-pole exclusion for the physical-scalar channels of B0. The maximum was

$$\max_{\sqrt{s}, i} |a_0^i| = 0.0207110$$

at 20 TeV, approaching the physical-subspace high-energy value 0.0207374. Analytic logarithmic angular integration agreed with independent 192-point Gauss-Legendre integration at representative energies; the largest absolute matrix difference was 2.8×10^{-9} .

MODEL-DEPENDENT RESULT

B0 passes the complete finite-energy **physical-scalar** tree-level test over the scanned interval. This is not yet the complete electroweak/gauge-sector unitarity result because external Goldstones and vectors and gauge-exchange diagrams have not been included.

11.48 5. Independently regenerated scalar one-loop RGEs

The scalar-only beta functions were derived in two independent ways:

1. contraction of the fully symmetric four-index quartic tensor;
2. extraction from $\frac{1}{2} \text{Tr}[(V_4'')^2]$.

The two results agree exactly as rational polynomials. With $\beta_\lambda = d\lambda/d \ln \mu$,

$$\begin{aligned} 16\pi^2 \beta_{\lambda_H} &= 24\lambda_H^2 + \lambda_{HA}^2 + \lambda_{HB}^2, \\ 16\pi^2 \beta_{\lambda_A} &= 20\lambda_A^2 + 2\lambda_{HA}^2 + \lambda_{AB}^2, \\ 16\pi^2 \beta_{\lambda_B} &= 20\lambda_B^2 + 2\lambda_{HB}^2 + \lambda_{AB}^2, \\ 16\pi^2 \beta_{\lambda_{HA}} &= 12\lambda_H\lambda_{HA} + 8\lambda_A\lambda_{HA} + 4\lambda_{HA}^2 + 2\lambda_{AB}\lambda_{HB}, \\ 16\pi^2 \beta_{\lambda_{HB}} &= 12\lambda_H\lambda_{HB} + 8\lambda_B\lambda_{HB} + 4\lambda_{HB}^2 + 2\lambda_{AB}\lambda_{HA}, \\ 16\pi^2 \beta_{\lambda_{AB}} &= 8\lambda_A\lambda_{AB} + 8\lambda_B\lambda_{AB} + 4\lambda_{AB}^2 + 4\lambda_{HA}\lambda_{HB}. \end{aligned}$$

The coefficients reproduce the decoupled $O(4)$ and $O(2)$ limits. Standard Model gauge and third-family Yukawa terms were then added in the one-loop running code. Dimensionful masses and cubic coefficients are not yet included.

11.49 6. Named running benchmarks

11.49.1 B0 — weak portals

$$(\lambda_{HA}, \lambda_{HB}) = (0.020, 0.030)$$

at $\mu_0 = 173 \text{ GeV}$. This is the same quartic point used in the tree-level vacuum and high-energy unitarity audits. The one-loop system gives

$$\mu_{BFB} = 3.5739 \times 10^8 \text{ GeV}, \quad \mu_U = 2.1109 \times 10^{15} \text{ GeV}, \quad \mu_{4\pi} = 3.9611 \times 10^{15} \text{ GeV}.$$

The first failure is the running Higgs quartic becoming negative. B0 therefore fails the chosen **absolute tree-level BFB-at-every-scale criterion**. This does not by itself prove an unacceptable vacuum lifetime; a metastability calculation using the RG-improved effective potential would be required for that different question.

11.49.2 B1 — diagnostic stabilizing portals

$$(\lambda_{HA}, \lambda_{HB}) = (0.18, 0.18),$$

with the other dimensionless initial values unchanged. No BFB failure occurs before perturbation theory fails. The minimum copositivity margin is 0.004086, while

$$\mu_U = 7.1790 \times 10^{14} \text{ GeV}, \quad \mu_{4\pi} = 1.3534 \times 10^{15} \text{ GeV}.$$

The estimates are stable under 3000, 6000, and 9000 logarithmic sample points. The ODE solver eventually stops after the theory has already crossed both declared perturbative boundaries; its later blow-up is not used as a physical prediction.

B1 is a theoretical diagnostic, **not yet a phenomenologically accepted benchmark**. In particular, changing the portals requires recomputing the dimensionful mass parameters if the same pole-mass targets are desired, and Higgs-portal direct-detection and invisible-width constraints must be imposed before claiming viability.

11.50 7. Referee attack

11.50.1 What has actually been shown

- The B0 scalar potential is bounded from below at the input scale.
- The B0 inert electroweak point is the global tree-level minimum under explicit sufficient inequalities.
- The full high-energy scalar/Goldstone quartic matrix is unitary at B0.
- The full finite-energy physical-scalar matrix is unitary for B0 from threshold to 20 TeV.
- Two independent scalar-only one-loop derivations agree exactly.
- Weak portals do not stabilize the running Higgs quartic in the one-loop approximation used.
- Stronger portals can postpone BFB failure beyond the perturbative domain, but they make the theory reach strong coupling earlier.

11.50.2 What has not been shown

- a complete finite-energy gauge-independent scalar/Goldstone/vector bound;
- two-loop or threshold-matched running;
- the running of masses and cubic coefficients;
- tunnelling lifetime or metastability;
- collider, relic-density, direct-detection, or cosmological viability of B0 or B1;
- ultraviolet completion;
- any new experimentally confirmed prediction.

11.50.3 Failure conditions

This benchmark branch fails if a complete calculation finds a deeper vacuum with unacceptable lifetime, a finite-energy unitarity violation below its intended cutoff, exclusion by portal/dark-sector data throughout the theoretically allowed region, or radiative generation of an operator forbidden by the claimed exact symmetry.

11.51 8. Status and next gate

Test	Status	Reason
Tree-level boundedness	Achieved for B0	Exact copositivity conditions pass
Global inert electroweak vacuum	Achieved for B0	Analytic sufficient proof plus numerical checks
High-energy scalar unitarity	Achieved for B0	Complete 36-channel quartic matrix
Scalar one-loop RGE derivation	Achieved	Two exact methods agree
One-loop absolute stability	Failed for B0; provisional for B1	B0 loses BFB; B1 reaches perturbative boundary first
Finite-energy physical-scalar unitarity	Provisional pass for B0	All scalar contacts and exchanges included to 20 TeV
Finite-energy gauge-sector unitarity	Open	Goldstone/vector channels and gauge diagrams incomplete
Full quantum consistency	Partial	Two-loop, thresholds, and dimensionful RGEs absent
Phenomenology	Open	No global experimental constraint analysis yet
TP-specific prediction	Open	The symmetry-protected null transition remains the current lead

The next defensible step is the gauge-complete finite-energy scattering audit followed by a deliberately minimal phenomenology benchmark. Grand-unification claims remain quarantined.

11.52 Reproducible files

- `code/tpd_ew_vacuum.py`
- `code/tpd_high_energy_unitarity.py`
- `code/tpd_finite_energy_unitarity.py`
- `code/tpd_scalar_rge.py`
- `code/tpd_quartic_rge_running.py`
- `code/test_tpd_phase3.py`
- `audits/tpd_ew_vacuum.json`
- `audits/tpd_high_energy_unitarity.json`
- `audits/tpd_finite_energy_unitarity.json`
- `audits/tpd_scalar_rge.json`
- `audits/tpd_quartic_rge_running.json`
- `audits/tpd_quartic_rge_B1.json`
- `audits/tpd_rge_convergence_B1.json`

11.53 S5 Ternary Polarity checkpoint 7

11.54 Dimensionful one-loop RGEs, portal trade-off, and preliminary metastability

Research status: internal technical audit

Theory version: TP-D v0.1, with v0.2 update recommended

Date: 19 September 2026

Epistemic level: exact scalar one-loop derivations; model-dependent running; leading-only metastability diagnostic

11.55 Executive verdict

The dimensionful scalar one-loop system has been derived twice and agrees exactly. The dual $Z_3 \times Z_3$ cubic structure is preserved: no forbidden mixed mass or mixed cubic is generated when the symmetry is exact.

The running analysis exposes a real tension:

- weak Higgs portals leave the Higgs quartic negative above approximately 3.57×10^8 GeV;
- stronger equal portals can keep all quartics bounded, but their mass-weighted contributions drive the running Higgs quadratic parameter through zero around 2.31×10^8 GeV;
- an asymmetric portal concentrated on the lighter species substantially improves this trade-off, reaching approximately 1.46×10^{15} GeV before the simple inert-vacuum certificate fails;
- the best scanned asymmetric point is near-critical and therefore not robust against omitted higher-order and threshold corrections.

The leading quartic-bounce estimate suggests that B0 is long-lived inside the declared perturbative interval. This is **not** a precision lifetime calculation because the most negative quartic occurs at the cutoff, not at a controlled stationary bounce scale.

11.56 1. Dimensionful normalization

The canonical potential contains

$$V \supset -\mu_H^2 H^\dagger H + m_A^2 |A|^2 + m_B^2 |B|^2 + \left(\frac{\mu_A}{3} A^3 + \frac{\mu_B}{3} B^3 + \text{h.c.} \right).$$

Independent rephasings choose the cubic terms to be negative along the phase-minimized positive radial directions. For

$$A = \frac{A_1 + iA_2}{\sqrt{2}},$$

the convention used by the tensor calculation is

$$V_A^{(3)} = -\frac{k_A}{3} A_1^3 + k_A A_1 A_2^2, \quad k_A = \frac{|\mu_A|}{\sqrt{2}},$$

and similarly for B . This differs from the common literature convention $\mu_3(S^3 + S^{\dagger 3})/2$; the conversion is $\mu_3 = 2\mu_A/3$. The distinction is purely normalization and must be retained when importing published bounds [Bel12, Hek19].

11.57 2. Independent derivations

The complete eight-real-scalar potential was evaluated in two ways.

11.57.1 Method A — one-loop effective-potential divergence

The scalar contribution to the one-loop beta-potential numerator was extracted from

$$\frac{1}{2} \text{Tr}[(V'')^2].$$

Coefficients of quadratic, cubic, and quartic field monomials were projected onto the canonical parameters.

11.57.2 Method B — general tensor contractions

For

$$V = \frac{1}{2}m_{ab}^2\phi_a\phi_b + \frac{1}{3!}h_{abc}\phi_a\phi_b\phi_c + \frac{1}{4!}\lambda_{abcd}\phi_a\phi_b\phi_c\phi_d,$$

the pure-scalar one-loop relations are

$$(16\pi^2)\beta_{m_{ab}^2} = \lambda_{abef}m_{ef}^2 + h_{aef}h_{bef},$$

$$(16\pi^2)\beta_{h_{abc}} = \lambda_{abef}h_{cef} + \lambda_{acef}h_{bef} + \lambda_{bcef}h_{aef}.$$

These agree with the general formulas of Luo, Wang, and Xiao, including the permutation normalization [Luo02b]. The two in-project implementations agree coefficient by coefficient using exact rational arithmetic.

11.58 3. Scalar-only dimensionful beta functions

With $\beta_x = dx/d\ln\mu$, the exact scalar contributions are

$$\begin{aligned} 16\pi^2\beta_{\mu_H^2}^{\text{scalar}} &= 12\lambda_H\mu_H^2 - 2\lambda_{HA}m_A^2 - 2\lambda_{HB}m_B^2, \\ 16\pi^2\beta_{m_A^2}^{\text{scalar}} &= 8\lambda_A m_A^2 + 2\lambda_{AB}m_B^2 - 4\lambda_{HA}\mu_H^2 + 4|\mu_A|^2, \\ 16\pi^2\beta_{m_B^2}^{\text{scalar}} &= 8\lambda_B m_B^2 + 2\lambda_{AB}m_A^2 - 4\lambda_{HB}\mu_H^2 + 4|\mu_B|^2, \\ 16\pi^2\beta_{\mu_A}^{\text{scalar}} &= 12\lambda_A\mu_A, \\ 16\pi^2\beta_{\mu_B}^{\text{scalar}} &= 12\lambda_B\mu_B. \end{aligned}$$

The Standard Model gauge and third-family Yukawa contribution added to the first equation is

$$\mu_H^2 \left[2(3y_t^2 + 3y_b^2 + y_\tau^2) - \frac{9}{2}g_2^2 - \frac{3}{2}g_1^2 \right].$$

The scalar quartic beta functions remain those independently verified in checkpoint 6.

11.58.1 Symmetry preservation

The alternative $A_1A_2^2$ and $B_1B_2^2$ tensor projections yield the same cubic beta functions as the A_1^3 and B_1^3 projections. Thus the exact Z_3 cubic tensors retain their required ratios. Because every retained vertex is neutral under both discrete factors, mixed bilinears and mixed cubics remain absent to all perturbative orders.

11.59 4. Running certificates

The one-loop integration now stops at the first 4π dimensionless-coupling boundary instead of continuing into numerical blow-up. It monitors:

- exact three-field copositivity;
- $\mu_H^2 > 0$, $m_A^2 > 0$, and $m_B^2 > 0$;
- nonnegative portal quartics for the analytic sufficient proof;
- $9\lambda_A m_A^2 - |\mu_A|^2 \geq 0$;
- $9\lambda_B m_B^2 - |\mu_B|^2 \geq 0$;
- the complete high-energy scalar-unitarity eigenvalue.

These running inequalities are diagnostics constructed from the RG-improved tree potential. At finite perturbative order they are not themselves scale-independent observables.

11.59.1 B0 — weak portals

$$(\lambda_{HA}, \lambda_{HB}) = (0.020, 0.030).$$

The interpolated first BFB failure is

$$\mu_{BFB} = 3.5707 \times 10^8 \text{ GeV}.$$

All mass and cubic sufficient-condition margins remain positive up to the later perturbative boundaries. Therefore the first B0 failure is specifically the negative Higgs quartic, not a dark tachyon or cubic-induced dark minimum.

11.59.2 B1 — equal stabilizing portals

$$(\lambda_{HA}, \lambda_{HB}) = (0.18, 0.18).$$

The quartic potential remains copositive over the scanned perturbative interval, but

$$\mu_H^2(\mu) = 0 \quad \text{at} \quad \mu \simeq 2.3054 \times 10^8 \text{ GeV}.$$

This does not prove vacuum decay. It means the simple all-scale analytic certificate derived for a negative Higgs quadratic term no longer applies. The result also quantifies a hierarchy/naturalness tension: the same portal terms that raise β_{λ_H} contribute negatively to $\beta_{\mu_H^2}$, weighted by m_A^2 and m_B^2 .

11.59.3 B2 — asymmetric diagnostic branch

The beta function of the Higgs quartic depends on

$$\lambda_{HA}^2 + \lambda_{HB}^2,$$

whereas the portal contribution to the Higgs quadratic beta function is weighted approximately by

$$\lambda_{HA} m_A^2 + \lambda_{HB} m_B^2.$$

At fixed stabilizing quartic norm, concentrating the portal on the lighter state reduces the mass-weighted correction. The diagnostic point

$$(\lambda_{HA}, \lambda_{HB}) = (0.2385, 0)$$

has no BFB failure before its perturbative boundary and retains the sufficient inert certificate until

$$\mu \simeq 1.4624 \times 10^{15} \text{ GeV}.$$

Its scalar-unitarity and 4π boundaries are approximately

$$\mu_U = 2.1555 \times 10^{15} \text{ GeV}, \quad \mu_{4\pi} = 4.0470 \times 10^{15} \text{ GeV}.$$

However, its minimum BFB margin is only

$$3.3 \times 10^{-6}.$$

The point is therefore near-critical and highly sensitive to threshold matching, two-loop terms, and input uncertainties. B2 is not promoted to a phenomenological benchmark.

11.60 5. Sparse portal scan

For equal portals between 0.02 and 0.22, increasing the portal first postpones the Higgs-quartic zero, then makes the Higgs quadratic sign change the earlier failure. At the fixed masses and self-couplings used here, the largest reach occurs near the crossover rather than at the largest portal.

An asymmetric scan confirms that the lighter-state portal performs substantially better. This is a model-building result, but not a universal theorem: changing dark masses, self-couplings, cubic terms, or adding threshold matching changes the optimum.

11.61 6. Leading metastability diagnostic

For a negative quartic potential, the leading Fubini action is

$$S_4 \simeq \frac{8\pi^2}{3|\lambda_H(\mu)|},$$

with a rough expected-event exponent

$$\ln N_{\text{decay}} \sim 4 \ln(T_U \mu) - S_4.$$

This approximation and its limitations are standard in electroweak-vacuum analyses [Isi01, Esp20].

For B0, restricting the calculation to the interval below the first scalar-unitarity boundary gives

$$\lambda_{H,\text{min}} = -0.027390, \quad S_4 \simeq 960.9, \quad \log_{10} N_{\text{decay}} \sim -188.7.$$

This leading diagnostic is compatible with a very long-lived vacuum. It is not a controlled lifetime prediction because:

1. the minimum occurs at the declared cutoff endpoint;
2. $\beta_{\lambda_H} \neq 0$ there, so the calculation has not selected a stationary bounce scale;
3. two-loop running, threshold matching, fluctuation determinants, gauge-consistent effective couplings, gravity, higher-dimensional operators, and multi-field bounces are absent.

B1 and B2 keep $\lambda_H \geq 0$ inside their declared perturbative intervals, so this particular negative-quartic bounce is absent there. That does not replace a complete multi-field vacuum analysis.

11.62 7. Referee attack and classification

Question	Result	Classification
Scalar mass RGEs	Two exact methods agree	Derived
Scalar cubic RGEs	Two exact methods agree	Derived
Preservation of each Z_3 cubic tensor	Verified	Derived
B0 absolute BFB at all scales	Fails near 3.57×10^8 GeV	Negative model result
B0 lifetime	Leading diagnostic appears safe	Provisional; UV sensitive
B1 quartic stability	Passes to perturbative boundary	Model-dependent
B1 inert-vacuum certificate	Fails via $\mu_H^2 = 0$	Certificate failure, not proven decay
B2 improved scale reach	Reaches 1.46×10^{15} GeV	Near-critical diagnostic
Threshold-matched running	Not done	Open
Two-loop uncertainty	Not done	Open
Full Coleman–Weinberg/multi-field bounce	Not done	Open
Gauge-origin contributions	Undefined until TP-G is specified	Blocked by model choice

11.63 8. Consequences for the theory programme

1. TP-D remains a consistent low-energy scalar EFT branch.
2. B0 should be described as metastability-compatible at leading order, not absolutely stable to high scales.
3. B1 should not be advertised as a clean solution to vacuum stability; it trades the quartic problem for a quadratic-parameter problem.
4. B2 demonstrates a repair mechanism through asymmetric portals but is too close to criticality to be trusted without higher-order work.
5. None of these results produces a TP-specific observable or proves novelty beyond known product-discrete scalar models.
6. A gauge-complete unitarity/anomaly audit cannot begin until the parent gauge field content and chiral charge assignments are frozen.

11.64 9. Reproducibility

- code/tpd_dimensionful_rge.py
- code/tpd_full_rge_running.py
- code/tpd_portal_tradeoff_scan.py
- code/tpd_metastability_diagnostic.py
- code/test_tpd_phase3.py
- audits/tpd_dimensionful_rge.json
- audits/tpd_full_rge_B0.json
- audits/tpd_full_rge_B1.json
- audits/tpd_full_rge_B2.json
- audits/tpd_full_rge_convergence.json
- audits/tpd_portal_tradeoff_scan.json
- audits/tpd_metastability_diagnostic.json

Thirteen automated regression tests pass at this checkpoint.

11.65 Literature keys

- [Luo02b] M. Luo, H. Wang, and Y. Xiao, *Physical Review D* **67**, 065019 (2003), DOI 10.1103/PhysRevD.67.065019.
- [Bel12] G. Bélanger et al., *JCAP* **2013**(01), 022, DOI 10.1088/1475-7516/2013/01/022.
- [Hek19] A. Hektor, A. Hryczuk, and K. Kannike, *JHEP* **03** (2019) 204, DOI 10.1007/JHEP03(2019)204.
- [Isi01] G. Isidori, G. Ridolfi, and A. Strumia, *Nuclear Physics B* **609**, 387–409 (2001), DOI 10.1016/S0550-3213(01)00302-9.
- [Esp20] J. R. Espinosa, *JCAP* **2020**(06), 052, DOI 10.1088/1475-7516/2020/06/052.

11.66 S6 Ternary Polarity checkpoint 8

11.67 Minimal gauge origin, anomaly audit, parent vacuum, and enlarged scalar unitarity

Research status: internal technical audit

Theory version: TP-G0, a candidate parent of TP-D v0.2

Date: 19 September 2026

Epistemic level: explicit model construction; model-dependent benchmark; phenomenology incomplete

11.68 Executive verdict

A minimal scalar-only gauge parent exists:

$$U(1)_m \times U(1)_E \longrightarrow Z_3^{(m)} \times Z_3^{(E)}.$$

It uses two charge-three Higgs fields and reproduces the exact low-energy TP-D charges and cubic interactions. Because no chiral fermion carries either new charge, all continuous gauge, mixed-gauge, and gauge-gravitational anomaly coefficients vanish trivially.

An explicit benchmark, TP-G0, has:

- a certified global tree-level vacuum in its factorized parameter slice;
- positive gauge kinetic terms;
- two massive vectors, two radial modes, and the intended inert A, B spectrum;
- a complete 78-channel scalar/Goldstone quartic matrix satisfying high-energy tree unitarity.

This is a **conditional pass for existence of a gauge origin**, not a complete UV completion. The parent is a conventional product-Abelian residual-discrete construction, introduces many new parameters, permits three Abelian kinetic mixings, produces local cosmic strings, and has not passed full threshold RG, finite-energy vector scattering, collider, precision, or cosmological tests.

The standalone TP-D high-scale running cannot be imported into TP-G0. The radial modes and new vectors enter near 0.63 and 0.90 TeV, respectively, so the beta functions change above those thresholds. All TP-D numbers at 10^8 – 10^{15} GeV remain conditional results for an EFT in which no such low-scale parent states have appeared; they are not predictions of TP-G0.

11.69 1. Frozen parent field content

$$G_{\text{parent}} = G_{SM} \times U(1)_m \times U(1)_E.$$

Field	Q_m	Q_E	Role
Φ_m	3	0	breaks $U(1)_m$ to $Z_3^{(m)}$
Φ_E	0	3	breaks $U(1)_E$ to $Z_3^{(E)}$
A	1	0	low-energy (1, 0) state
B	0	1	low-energy (0, 1) state

All four fields are complex Lorentz scalars and Standard-Model singlets. Standard-Model fermions and the Higgs doublet are neutral under the two new factors.

The covariant derivatives include

$$D_\mu \Phi_m = (\partial_\mu + 3ig_m X_\mu^m) \Phi_m, \quad D_\mu A = (\partial_\mu + ig_m X_\mu^m) A,$$

and the analogous E expressions.

11.70 2. Residual group

Let

$$\langle \Phi_m \rangle = \frac{f_m}{\sqrt{2}}, \quad \langle \Phi_E \rangle = \frac{f_E}{\sqrt{2}}.$$

Vacuum-preserving gauge transformations satisfy

$$e^{3i\alpha_m} = e^{3i\alpha_E} = 1.$$

Therefore

$$\alpha_m = \frac{2\pi n_m}{3}, \quad \alpha_E = \frac{2\pi n_E}{3},$$

with $n_m, n_E \in \{0, 1, 2\}$. The surviving nine transformations form

$$Z_3^{(m)} \times Z_3^{(E)},$$

acting on A and B exactly as in TP-D. This is the standard residual-discrete gauge mechanism [Kra89, Ban10].

11.71 3. Complete renormalizable scalar operator basis

Define

$$\mathcal{S} = \{\Phi_m, \Phi_E, A, B\}.$$

The general gauge-invariant scalar potential contains:

1. one quadratic modulus for every $S_i \in \mathcal{S}$;
2. one self-quartic $|S_i|^4$ for every field;
3. all six pairwise modulus quartics $|S_i|^2 |S_j|^2$;
4. four Higgs portals $(H^\dagger H) |S_i|^2$;
5. the two phase-sensitive structures

$$\kappa_m \Phi_m^\dagger A^3 + \text{h.c.}, \quad \kappa_E \Phi_E^\dagger B^3 + \text{h.c.}.$$

No gauge-invariant scalar monomial of engineering dimension one or three exists. Exhaustive enumeration found four quadratic monomials and fourteen degree-four monomials when Hermitian-conjugate phase-sensitive terms are counted separately.

After symmetry breaking,

$$\frac{\kappa_m f_m}{\sqrt{2}} A^3 + \text{h.c.} = \frac{\mu_A}{3} A^3 + \text{h.c.},$$

so

$$\mu_A = \frac{3\kappa_m f_m}{\sqrt{2}}, \quad \mu_B = \frac{3\kappa_E f_E}{\sqrt{2}}.$$

The low-energy cubic is therefore matched rather than inserted by hand.

11.72 4. Gauge kinetic mixing correction

The most general Abelian kinetic sector includes

$$-\frac{1}{4}F_Y^2 - \frac{1}{4}F_m^2 - \frac{1}{4}F_E^2 - \frac{\epsilon_{Ym}}{2}F_Y F_m - \frac{\epsilon_{YE}}{2}F_Y F_E - \frac{\epsilon_{mE}}{2}F_m F_E.$$

Omitting the two hypercharge mixings would be inconsistent with the most-general-action rule. Kinetic mixing is a standard allowed interaction between Abelian factors [Hol86].

In the frozen scalar-only charge assignment,

$$\text{Tr}(YQ_m) = \text{Tr}(YQ_E) = \text{Tr}(Q_m Q_E) = 0,$$

because no field carries both relevant charges. Thus zero mixing has no one-loop matter source. It is not enforced by the gauge group itself. The TP-G0 zero-mixing slice additionally imposes independent charge-conjugation automorphisms C_m and C_E , under which X_μ^m or X_μ^E changes sign and the corresponding charged scalars are conjugated. With real scalar couplings, these symmetries protect all new-sector kinetic mixings from perturbative generation.

Without those extra automorphisms, all three ϵ 's are independent parameters and must be constrained experimentally.

11.73 5. Gauge spectrum

When hypercharge mixing vanishes, the unmixed masses are

$$M_m^2 = 9g_m^2 f_m^2, \quad M_E^2 = 9g_E^2 f_E^2.$$

For nonzero $\epsilon_{mE} = \epsilon$, canonical normalization gives

$$M_\pm^2 = \frac{M_m^2 + M_E^2 \pm \sqrt{(M_m^2 - M_E^2)^2 + 4\epsilon^2 M_m^2 M_E^2}}{2(1 - \epsilon^2)}.$$

The kinetic matrix requires $|\epsilon| < 1$ on this two-factor slice. A 500-point numerical generalized-eigenvalue check reproduced the analytic formula to maximum relative error 8.3×10^{-14} .

The phases of Φ_m and Φ_E are eaten. Two physical radial modes remain.

11.74 6. Anomaly audit

Gauge anomalies receive contributions from chiral fermions, not scalars. TP-G0 contains no fermion charged under either new $U(1)$, while all Standard-Model chiral fermions are neutral. Consequently,

$$[U(1)_m]^3 = [U(1)_E]^3 = 0,$$

all mixed Abelian coefficients vanish, all mixed Standard-Model coefficients vanish, and both gravitational- $U(1)$ coefficients vanish.

The continuous parent is therefore anomaly-free in this restricted field content, and its residual discrete subgroup inherits a consistent gauge origin. This result does **not** survive automatically if chiral portal matter or Standard-Model charges are added; every such extension requires a new audit [Iba91].

11.75 7. TP-G0 benchmark

Choose

$$f_m = f_E = 1 \text{ TeV}, qquad g_m = g_E = 0.30, qquad \lambda_{\Phi_m} = \lambda_{\Phi_E} = 0.20,$$

with the B0 low-energy quartics and masses. Set the other modulus portals and all kinetic mixings to zero on the charge-conjugation-symmetric slice. Cubic matching fixes

$$\kappa_m = 0.0282843, qquad \kappa_E = 0.0424264.$$

The tree spectrum is

$$m_{\rho_m} = m_{\rho_E} = 632.46 \text{ GeV}, \quad M_m = M_E = 900 \text{ GeV},$$

plus $M_A = 180 \text{ GeV}$, $M_B = 400 \text{ GeV}$, and the Standard-Model-like Higgs.

11.75.1 Parent BFB and global vacuum

For either (Φ, A) sector with vanishing modulus portal, the phase-minimized quartic along $t = |\Phi|/|A|$ is proportional to

$$f_4(t) = \lambda_\Phi t^4 + \lambda_A - 2|\kappa|t.$$

Its minimum occurs at

$$t_* = \left(\frac{|\kappa|}{2\lambda_\Phi} \right)^{1/3},$$

with margin

$$f_4(t_*) = \lambda_A - \frac{3}{2}|\kappa|t_*.$$

The TP-G0 margins are 0.18246 and 0.21988, respectively.

Interior stationary points with dark radial value $a > 0$ reduce to positive roots of

$$4\lambda_\Phi M(x) [4M(x)^2 - 9\kappa^2 f^2 x] - 27\kappa^4 x^3 = 0,$$

where $x = a^2$ and $M(x) = m_A^2 + \lambda_A x$. Both benchmark polynomials have no positive real root. Boundary directions and infinity are positive, so each factorized sector has its global minimum at $(|\Phi|, |A|) = (f/\sqrt{2}, 0)$ in complex-field notation. Nonnegative H - A , H - B , and A - B portals complete the benchmark global-vacuum certificate.

The proof applies to this factorized parameter slice. The zero modulus portals are not claimed to form a closed RG surface.

11.76 8. Enlarged high-energy scalar matrix

The parent contains twelve real scalar components and 78 normalized two-particle channels. The complete scalar-potential quartic tensor includes the phase-sensitive $\kappa_m \Phi_m^\dagger A^3$ and $\kappa_E \Phi_E^\dagger B^3$ interactions.

For TP-G0,

$$\max |\Lambda| = 1.05056, \quad \max |a_0| = 0.0209001,$$

well below $|\operatorname{Re} a_0| \leq 1/2$. The 36-channel $H + A + B$ principal submatrix reproduces the independent low-energy generator exactly.

This is a scalar/Goldstone quartic result. Gauge exchanges, transverse vectors, finite-energy diagrams, and nonzero kinetic mixing remain outside it.

11.77 9. Topological defects

Each breaking $U(1) \rightarrow Z_3$ has vacuum manifold $U(1)/Z_3$ with

$$\pi_1(U(1)/Z_3) = \mathbb{Z},$$

so local cosmic strings are expected. Their flux is quantized in units proportional to $2\pi/(3g)$. For $f = 1 \text{ TeV}$, a tension estimate $\mu_{string} \sim 2\pi f^2$ gives a gravitational strength of order

$$G\mu_{string} \sim 4 \times 10^{-32}$$

using the reduced Planck-mass convention. This is cosmologically negligible as a gravitational source, though the particle-emission history has not been calculated.

There are no physical domain walls while the residual discrete gauge group remains unbroken. A later A or B vacuum expectation value would require a new string-wall analysis.

11.78 10. Occam and novelty audit

TP-G0 solves one problem: it realizes the exact product Z_3 as a residual gauge symmetry and removes the need to postulate an exact global discrete symmetry. Its cost is substantial:

- two gauge couplings;
- two breaking scales;
- two radial self-couplings and multiple modulus portals;
- two matching couplings κ_m, κ_E ;
- up to three Abelian kinetic mixings;
- two new vectors and two new radial modes;
- local cosmic strings.

The construction is not novel by itself. Residual discrete gauge symmetries, Abelian kinetic mixing, and product-discrete dark sectors are established structures. Any TP novelty must come from a restricted relation or observable not already present in generic versions of these models.

11.79 11. Gate status

Requirement	Status
Explicit parent gauge group and charges	Achieved
Derivation of residual $Z_3 \times Z_3$	Achieved
Complete renormalizable scalar operator enumeration	Achieved
Continuous anomaly cancellation	Achieved trivially for scalar-only matter
Discrete gauge consistency	Inherited conditionally from parent
General kinetic mixing included	Achieved
Positive-kinetic benchmark	Achieved on protected zero-mixing slice
Global tree vacuum	Achieved for TP-G0 factorized slice
High-energy scalar/Goldstone unitarity	Achieved for TP-G0
Full gauge-sector finite-energy unitarity	Open
Parent one-/two-loop RGEs and threshold matching	Open
Collider and precision viability	Open
Cosmic-string particle/cosmology analysis	Open
Parameter reduction or unique prediction	Failed to emerge at this stage

The correct verdict is **conditional pass as a minimal anomaly-free gauge origin**, not UV completion and not unification.

In particular, the B0/B1/B2 ultraviolet scales quoted in checkpoint 7 must not be combined with TP-G0 until matching at the (h_0 , h_0 , X_m , X_E) thresholds and the parent RG system have been calculated.

11.80 12. Reproducibility

- `code/tpd_gauge_origin_audit.py`
- `code/tpd_gauge_vacuum.py`
- `code/tpd_gauge_high_energy_unitarity.py`
- `audits/tpd_gauge_origin_audit.json`
- `audits/tpd_gauge_vacuum.json`
- `audits/tpd_gauge_high_energy_unitarity.json`

Seventeen automated regression tests pass after integrating this checkpoint.

11.81 Literature keys

- [Kra89] L. M. Krauss and F. Wilczek, *Physical Review Letters* **62**, 1221–1223 (1989), DOI 10.1103/PhysRevLett.62.1221.
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11.82 S7 Ternary Polarity checkpoint 9

11.83 Parent scalar RG reconstruction and threshold-separated running

Research status: internal technical audit

Theory version: TP-G0 matched to TP-D v0.2

Date: 19 September 2026

Epistemic level: exact one-loop scalar algebra plus model-dependent leading running

11.84 Executive verdict

The general CP-conserving TP-G scalar potential has seventeen real dimensionless couplings: five self quartics, ten modulus portals, and two phase-sensitive quartics. Its complete scalar-only one-loop beta functions have been derived with exact rational arithmetic. Representative components were independently regenerated from the four-index quartic tensor, and the TP-D subsystem is recovered exactly.

The factorized TP-G0 benchmark is **not an RG-invariant surface**. Even when

$$\lambda_{A\Phi_m} = \lambda_{B\Phi_E} = 0$$

at the matching scale, scalar and gauge loops give

$$\begin{aligned} 16\pi^2 \beta_{\lambda_{A\Phi_m}} \Big|_0 &= 36\kappa_m^2 + 108g_m^4, \\ 16\pi^2 \beta_{\lambda_{B\Phi_E}} \Big|_0 &= 36\kappa_E^2 + 108g_E^4. \end{aligned}$$

This invalidates extrapolation of the factorized tree-vacuum proof away from the matching scale. It does not immediately invalidate the benchmark: the generated portals are positive, and a sufficient five-field boundedness certificate survives until the Higgs quartic becomes negative. A general running-vacuum analysis is nevertheless required.

11.85 1. Exact parent scalar system

For

$$\{H, A, B, \Phi_m, \Phi_E\},$$

the scalar potential contains

$$5 \text{ self quartics} + 10 \text{ modulus portals} + 2 \text{ phase-sensitive quartics} = 17$$

real physical couplings after the two isolated phases of κ_m, κ_E are removed by field redefinitions. The scalar-only one-loop potential was computed as

$$16\pi^2 \beta_V^{(1)} = \frac{1}{2} \text{Tr}[(V_4'')^2].$$

Independent quartic-tensor contractions checked $\lambda_H, \lambda_A, \lambda_{H\Phi_m}, \lambda_{A\Phi_m}, \lambda_{\Phi_m\Phi_E}, \kappa_m, \kappa_E$ exactly. The four real-component projections of each phase-sensitive operator yield the same beta function, which is a nontrivial covariance check.

Selected results are

$$\begin{aligned} 16\pi^2 \beta_{\lambda_A}^{scalar} &= 20\lambda_A^2 + 2\lambda_{HA}^2 + \lambda_{AB}^2 + \lambda_{A\Phi_m}^2 + \lambda_{A\Phi_E}^2 + 18\kappa_m^2, \\ 16\pi^2 \beta_{\lambda_{A\Phi_m}}^{scalar} &= 4\lambda_{A\Phi_m}^2 + 8(\lambda_A + \lambda_{\Phi_m})\lambda_{A\Phi_m} + 36\kappa_m^2 \\ &\quad + 2\lambda_{AB}\lambda_{B\Phi_m} + 2\lambda_{A\Phi_E}\lambda_{\Phi_m\Phi_E} + 4\lambda_{HA}\lambda_{H\Phi_m}, \\ 16\pi^2 \beta_{\kappa_m}^{scalar} &= \kappa_m(12\lambda_A + 6\lambda_{A\Phi_m}), \end{aligned}$$

with the (E) sector obtained by interchange. The complete expressions are stored in `audits/tpd_parent_scalar_rge.json`.

11.86 2. New-Abelian terms

On the charge-conjugation-protected zero-mixing slice,

$$16\pi^2\beta_{g_m} = \frac{10}{3}g_m^3, \quad 16\pi^2\beta_{g_E} = \frac{10}{3}g_E^3.$$

Each coefficient follows from one charge-one and one charge-three complex scalar:

$$b = \frac{1}{3}(1^2 + 3^2) = \frac{10}{3}.$$

Examples of exact gauge contributions are

$$\begin{aligned} 16\pi^2\beta_{\lambda_{A\Phi_m}}^{\text{gauge}} &= -60g_m^2\lambda_{A\Phi_m} + 108g_m^4, \\ 16\pi^2\beta_{\lambda_{\Phi_m}}^{\text{gauge}} &= -108g_m^2\lambda_{\Phi_m} + 486g_m^4, \\ 16\pi^2\beta_{\kappa_m}^{\text{gauge}} &= -36g_m^2\kappa_m. \end{aligned}$$

The analogous (E)-sector formulas follow by exchange. There is no one-loop inhomogeneous kinetic-mixing source because no active field is bi-charged. The imposed $C_m \times C_E$ automorphism protects the zero-mixing slice to all perturbative orders.

11.87 3. Threshold separation

The TP-D running of checkpoints 6 and 7 is a standalone-EFT calculation. It cannot be used above the TP-G0 radial and vector masses. For the benchmark,

$$m_{\rho_m} = m_{\rho_E} = 632.46 \text{ GeV}, \quad M_m = M_E = 900 \text{ GeV}.$$

The present leading calculation uses a common step matching scale of 1 TeV. It evolves TP-D from 173 GeV to that scale, turns on the parent fields, and then runs all 17 scalar couplings together with the SM and new gauge couplings. It neglects finite matching corrections and the physical splitting between the four thresholds.

11.88 4. TP-G0 running result

At 1 TeV, the running values inherited from the low theory include

$$\lambda_H = 0.09899, \quad \lambda_A = 0.20943, \quad \lambda_B = 0.26485,$$

$$\lambda_{HA} = 0.02142, \quad \lambda_{HB} = 0.03224, \quad \lambda_{AB} = 0.10470.$$

The parent inputs are $g_m = g_E = 0.30$, $\lambda_{\Phi_m} = \lambda_{\Phi_E} = 0.20$, $\kappa_m = 0.028284$, and $\kappa_E = 0.042426$.

The leading one-loop trajectory gives:

Diagnostic	Scale
Sufficient parent BFB certificate first fails	$3.56 \times 10^8 \text{ GeV}$
High-energy scalar/Goldstone unitarity boundary	$3.33 \times 10^{16} \text{ GeV}$
First (4) dimensionless-coupling boundary	$6.28 \times 10^{16} \text{ GeV}$

The BFB failure is the Higgs direction, as in the standalone B0 branch. The new Abelian gauge couplings do not become large first.

At 10 TeV, radiatively generated portals include

$$\lambda_{A\Phi_m} = 0.01311, \quad \lambda_{B\Phi_E} = 0.01368,$$

$$\lambda_{A\Phi_E} = 2.14 \times 10^{-5}, \quad \lambda_{B\Phi_m} = 2.05 \times 10^{-5},$$

and small Higgs-parent portals are generated at the next linked order. Thus the general operator basis becomes dynamically relevant even when many couplings are zero at matching.

11.89 5. What has and has not been shown

11.89.1 Derived

- The complete scalar-only one-loop beta system for all 17 parent quartics.
- Exact scalar and gauge sources that destroy the factorized zero-portal surface.
- Exact one-loop Abelian coefficients on the protected zero-mixing slice.
- A leading step-matched parent trajectory and its scalar perturbative boundaries.

11.89.2 Model-dependent

- The numerical 3.56×10^8 , 3.33×10^{16} , and 6.28×10^{16} GeV scales.
- The portal values quoted at 10 TeV.

11.89.3 Not shown

- Dimensionful parent running and a global running-vacuum proof.
- Finite one-loop threshold matching at each physical mass.
- Two-loop stability of the quoted scales.
- Full finite-energy vector/scalar unitarity.
- Collider, precision, relic-density, or direct-detection viability.
- A nonzero TP-specific prediction.

11.90 6. Referee classification

Finding	Classification	Consequence
Standalone TP-D high-scale flow was being juxtaposed with a TeV parent	Repairable dependency error	The two flows are now explicitly separated
Factorized parent portal zeros run	Model-dependent but mandatory correction	Replace the factorized proof by a general parent vacuum audit above matching
Generated portals are positive in TP-G0	Conditional survival	No immediate high-field instability from those portals
Higgs quartic becomes negative	Model-dependent warning	Metastability/UV sensitivity remains; no absolute-stability claim
Scalar perturbativity eventually fails	EFT cutoff	No extrapolation beyond the first controlled boundary

11.91 7. Reproducibility

- `code/tpd_parent_scalar_rge.py`
- `code/tpd_parent_gauge_rge.py`

- `code/tpd_parent_rge_running.py`
- `audits/tpd_parent_scalar_rge.json`
- `audits/tpd_parent_gauge_rge.json`
- `audits/tpd_parent_rge_running.json`

Twenty automated regression tests pass after this checkpoint.

11.92 Literature basis

The tensor normalization follows the general one-loop renormalization-group formulas of Luo, Wang, and Xiao [Luo02b]. Residual gauge symmetry and Abelian kinetic-mixing context are as cited in checkpoint 8 [Kra89, Hol86]. These references support the methods and known mechanisms; they do not establish TP-G0 phenomenology or novelty.

11.93 S8 Ternary Polarity checkpoint 11

11.94 Coupled TP-D freeze-out equations and rejection of B0 as a thermal-relic benchmark

Research status: internal technical audit

Theory version: exact TP-D

Date: 19 September 2026

Epistemic level: exact collision structure; leading tree-level numerical diagnostic

11.95 Executive verdict

The two-component collision system has been written in a convention that tracks one yield per charge state and explicitly includes annihilation, semi-annihilation, and $B\bar{B} \leftrightarrow A\bar{A}$ conversion. It passes detailed-balance and dark-particle-number conservation tests.

Using tree-level threshold cross sections, fixed $g_* = g_{*s} = 90$, and the stored B0 parameters gives

$$\Omega_A h^2 = 1.248, \quad \Omega_B h^2 = 0.572, \quad \boxed{\Omega_{TP} h^2 = 1.820}.$$

The reference cosmological value is $\Omega_c h^2 = 0.120 \pm 0.001$ in base Λ CDM [Pla20]. B0 therefore overcloses the Universe by a factor of about 15 in this leading calculation. The discrepancy remains large under deliberately favourable rate rescalings. B0 is consequently **rejected as a standard thermal-relic benchmark**. This is a benchmark failure, not a falsification of the TP-D symmetry or of every TP-D parameter point.

11.96 1. Yield convention and exact collision structure

Assume a symmetric plasma,

$$Y_A = Y_{\bar{A}}, \quad Y_B = Y_{\bar{B}},$$

where each $Y_i = n_i/s$ is the yield of one charge state. The total late-time energy density contains an explicit factor of two for particle plus antiparticle.

With $x = m_A/T$, the equations are

$$\begin{aligned} \frac{dY_A}{dx} &= -\frac{s}{Hx} \left[\langle \sigma v \rangle_{A\bar{A} \rightarrow SM} (Y_A^2 - Y_{A,eq}^2) + \frac{1}{2} \langle \sigma v \rangle_{AA \rightarrow \bar{A}h} (Y_A^2 - Y_A Y_{A,eq}) - C_{B \rightarrow A} \right], \\ \frac{dY_B}{dx} &= -\frac{s}{Hx} \left[\langle \sigma v \rangle_{B\bar{B} \rightarrow SM} (Y_B^2 - Y_{B,eq}^2) + \frac{1}{2} \langle \sigma v \rangle_{BB \rightarrow \bar{B}h} (Y_B^2 - Y_B Y_{B,eq}) + C_{B \rightarrow A} \right], \end{aligned}$$

where

$$C_{B \rightarrow A} = \langle \sigma v \rangle_{B\bar{B} \rightarrow A\bar{A}} \left[Y_B^2 - \left(\frac{Y_{B,eq}}{Y_{A,eq}} \right)^2 Y_A^2 \right].$$

The factor 1/2 in each semi-annihilation term agrees with the standard Z_3 collision equation [Bel12, Bel13]. Detailed balance makes every collision polynomial vanish at equilibrium. Conversion enters with opposite signs, so it cancels from the equation for $Y_A + Y_B$: it redistributes dark particle number but does not itself reduce it. These properties are exact for the stated processes and convention.

11.97 2. Tree-level threshold rates

For the B0 parameters,

$$m_A = 180 \text{ GeV}, \quad m_B = 400 \text{ GeV}, \quad \lambda_{HA} = 0.020, \quad \lambda_{HB} = 0.030, \quad \lambda_{AB} = 0.10,$$

$$\mu_A = 60 \text{ GeV}, \quad \mu_B = 90 \text{ GeV},$$

the threshold calculation gives:

Process class	$\langle\sigma v\rangle [\text{GeV}^{-2}]$	$\langle\sigma v\rangle [\text{cm}^3 \text{s}^{-1}]$
$A\bar{A} \rightarrow \text{SM}, \text{ including } hh$	2.60×10^{-10}	3.03×10^{-27}
$AA \rightarrow \bar{A}h$	8.45×10^{-11}	9.86×10^{-28}
$B\bar{B} \rightarrow \text{SM}, \text{ including } hh$	1.25×10^{-10}	1.46×10^{-27}
$BB \rightarrow \bar{B}h$	3.32×10^{-12}	3.88×10^{-29}
$B\bar{B} \rightarrow A\bar{A}$	5.55×10^{-10}	6.48×10^{-27}

The Higgs-mediated Standard-Model rate is computed using the tree-level off-shell Higgs width. The hh final state includes the contact, s -channel, and dark-scalar exchange diagrams. The semi-annihilation rate uses the cubic and Higgs-portal vertices in the canonical potential. All rates are evaluated at threshold rather than thermally averaged.

11.98 3. Numerical method

The system is integrated with an implicit Radau method in $\ln x$:

- $x \in [1, 10^4]$;
- relative tolerance 2×10^{-8} ;
- absolute tolerance 10^{-15} ;
- constant $g_* = g_{*s} = 90$;
- equilibrium densities use the exact Bessel function $K_2(m_i/T)$.

An independent implementation evolves total particle-plus-antiparticle yields with a BDF solver and separately reconstructed number-counting factors. It gives $\Omega_{TP}h^2 = 1.82015$, differing from the primary result by 2.6×10^{-5} fractionally. This validates the yield convention and ODE normalization; it does not independently validate the matrix elements.

The late yields are

$$Y_A = 1.2646 \times 10^{-11}, \quad Y_B = 2.6065 \times 10^{-12}$$

per charge state. The conversion process substantially reduces the heavy component, but it transfers population into A ; it cannot compensate for insufficient number-changing rates.

11.99 4. Robustness diagnostics

Diagnostic	$\Omega_{TP}h^2$
Nominal	1.820
$g_* = 80$	1.936
$g_* = 100$	1.722
Stop at $x = 10^3$	1.847
Stop at $x = 10^5$	1.817
Remove conversion	4.056
Remove semi-annihilation	2.002
Multiply all number-changing rates by 2	1.139

Diagnostic	$\Omega_{TP}h^2$
Multiply all number-changing rates by 10	0.364

The simple global-vacuum certificate permits approximately

$$|\mu_A| < 239 \text{ GeV}, \quad |\mu_B| < 598 \text{ GeV}$$

at the stored masses and quartics. Raising both cubics to 99% of those bounds still gives

$$\Omega_{TP}h^2 \simeq 0.980.$$

Thus the fixed B0 masses, portals, and quartics cannot be rescued merely by increasing the cubic couplings while remaining inside this certificate.

11.100 5. What is exact and what is provisional

11.100.1 Exact within the frozen reaction set

- the charge selection rules;
- the form of the collision polynomials;
- the 1/2 semi-annihilation number-counting factor;
- detailed balance;
- cancellation of conversion in total dark-particle number;
- the conclusion that conversion alone cannot cure overproduction.

11.100.2 Leading/model-dependent

- all numerical cross sections;
- the use of threshold rather than thermal averages;
- constant g_* and g_{*S} ;
- tree-level Standard-Model widths;
- neglect of thermal masses, kinetic decoupling, and higher-order corrections;
- the assumption of a standard radiation-dominated thermal history.

An independent matrix-element implementation and proper thermal averaging are still required before a precision viable benchmark can be certified. They are not expected to bridge the factor-15 nominal discrepancy; the explicit factor-ten stress test is included to make that inference transparent.

11.101 6. Consequences for the research programme

1. B0 remains useful as a vacuum/RGE/unitarity regression point, but it must no longer be called a viable dark-matter benchmark.
2. Direct-detection fraction rescaling cannot rescue an overclosing thermal point. The direct-detection likelihood should be applied only after a new point reproduces the observed total abundance or is explicitly treated as a subcomponent under a different production history.
3. The next TP-D task is a constrained scan over masses, portals, cubics, quartics, and conversion strength with vacuum and unitarity cuts applied before relic and LZ cuts.
4. The frozen TP-G0 parent is still separately blocked by its unintended ρ_E relic.

11.102 7. Failure statement

Under a standard radiation-dominated thermal history and the canonical B0 parameters, B0 fails if its thermally averaged and independently regenerated relic abundance remains above the measured dark-matter density. The present leading result exceeds it by enough that B0 is rejected now for benchmark use; an independent implementation is retained as a verification requirement, not as a reason to continue advertising the point as viable.

11.103 8. Reproducibility

- `code/tpd_coupled_relic.py`
- `audits/tpd_coupled_relic.json`
- `code/tpd_coupled_relic_crosscheck.py`
- `audits/tpd_coupled_relic_crosscheck.json`
- `code/test_tpd_phase3.py`

Twenty-seven automated regression tests pass after this checkpoint.

11.104 Verified literature keys

- [Bel12] G. Bélanger, K. Kannike, A. Pukhov, and M. Raidal, *JCAP* **04** (2012) 010, DOI 10.1088/1475-7516/2012/04/010, arXiv:1202.2962.
- [Bel13] G. Bélanger, K. Kannike, A. Pukhov, and M. Raidal, *JCAP* **01** (2013) 022, DOI 10.1088/1475-7516/2013/01/022, arXiv:1211.1014.
- [Esc14] S. Esch, M. Klasen, and C. E. Yaguna, *JHEP* **09** (2014) 108, DOI 10.1007/JHEP09(2014)108, arXiv:1406.0617.
- [Pla20] N. Aghanim et al. (Planck Collaboration), *Astronomy & Astrophysics* **641** (2020) A6, DOI 10.1051/0004-6361/201833910, arXiv:1807.06209.

11.105 S9 TP checkpoint 12 — direct-detection gate

Authoritative continuation state: 2026-09-19

Gate verdict: all four relic-compatible points from the 160-point leading-order scan are robustly excluded by LZ under the stated standard SI/local-density assumptions. This rejects the four points, not all TP-D parameter space.

11.106 Experimental input

- Source: J. Aalbers et al. (LZ Collaboration), *Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment*, Phys. Rev. Lett. **135**, 011802 (2025), [arXiv:2410.17036v3](https://arxiv.org/abs/2410.17036v3), version dated 2025-07-01.
- Curve: combined WS2024+WS2022, 4.2 tonne-years / 280 live days, spin-independent WIMP–nucleon power-constrained 90% C.L. upper limit.
- Data route: programmatic log–log interpolation of the solid-black vector path embedded in Figure 5. HEPData record 155182 remained blocked and was not retrieved.
- Cross-check: interpolation gives $2.18 \times 10^{-48} \text{ cm}^2$ at 40 GeV versus $2.2 \times 10^{-48} \text{ cm}^2$ stated in the paper.

11.107 Preserved parameter vectors

All masses and cubics are in GeV; bare mass parameters are in GeV^2 .

- **P151:** mA=287.3881512, mB=451.8144615, mu_A=552.4374723, mu_B=278.8235522, lambda_HA=0.09848361249, lambda_HB=0.2467728786, lambda_AB=0.01424436406, lambda_A=0.5697719149, lambda_B=0.2161664158, cubic_fraction_A=0.8646424330, cubic_fraction_B=0.4507755869, bare_mA2=79606.70000, bare_mB2=196656.09251.
- **P82:** mA=172.0264639, mB=351.5807483, mu_A=205.2434277, mu_B=483.6417227, lambda_HA=0.1040147876, lambda_HB=0.004932664608, lambda_AB=0.2436293520, lambda_A=0.2373680250, lambda_B=0.5665194537, cubic_fraction_A=0.8635839963, cubic_fraction_B=0.6095829745, bare_mA2=26440.19303, bare_mB2=123459.50295.
- **P141:** mA=141.5633314, mB=414.3938204, mu_A=53.73663092, mu_B=762.7198088, lambda_HA=0.1467871706, lambda_HB=0.1275889494, lambda_AB=0.007965433064, lambda_A=0.1138031249, lambda_B=0.4495388445, cubic_fraction_A=0.4252444761, cubic_fraction_B=0.9255360701, bare_mA2=15590.74292, bare_mB2=167854.74376.
- **P10:** mA=321.2231638, mB=343.4716475, mu_A=550.7208237, mu_B=983.4851594, lambda_HA=0.08403040257, lambda_HB=0.2326296108, lambda_AB=0.005419119393, lambda_A=0.4386327313, lambda_B=0.9848090620, cubic_fraction_A=0.8737365383, cubic_fraction_B=0.9918891067, bare_mA2=100637.17928, bare_mB2=110921.27034.

11.108 Component results

$$\xi_i = \frac{\Omega_i h^2}{0.1200}, \quad \sigma_{iN}^{\text{eff}} = \xi_i \sigma_{iN}^{\text{SI}}, \quad R_i = \frac{\sigma_{iN}^{\text{eff}}}{\sigma_{\text{LZ}}(m_i)}.$$

Point	Component m_i (GeV)		$\Omega_i h^2$	ξ_i	raw σ_i^{SI}	effective	LZ limit	R_i	Verdict
					(cm^2)	$\xi_i \sigma_i$			
P151	A	287.388	0.0565070	0.470892	1.025e-45	4.825e-46	9.011e-48	53.55	excluded
P151	B	451.814	0.0634679	0.528899	2.609e-45	1.380e-45	1.375e-47	100.38	excluded
P82	A	172.026	0.0271190	0.225992	3.176e-45	7.178e-46	5.291e-48	135.64	excluded
P82	B	351.581	0.1004482	0.837068	1.720e-48	1.439e-48	1.072e-47	0.134	not individually excluded
P141	A	141.563	0.0181996	0.151663	9.319e-45	1.413e-45	4.532e-48	311.85	excluded
P141	B	414.394	0.1119164	0.932637	8.288e-46	7.730e-46	1.260e-47	61.34	excluded

Point	Component	m_i (GeV)	$\Omega_i h^2$	ξ_i	raw σ_i^{SI} (cm ²)	effective $\xi_i \sigma_i$ (cm ²)	LZ limit (cm ²)	R_i	Verdict
P10	A	321.223	0.1240571	1.033809	5.975e-46	6.177e-46	9.918e-48	62.28	excluded; also in- dividu- ally over- abun- dant in the lead- ing relic ap- proxi- mation
P10	B	343.472	0.0116976	0.097480	4.007e-45	3.906e-46	1.051e-47	37.17	excluded

Point verdicts: **P151 excluded; P82 excluded by A; P141 excluded; P10 excluded.** The smallest excluding ratio is 37.17.

11.109 Interpretation audit

- **Abundance rescaling:** each predicted component abundance is divided by the observed 0.1200, not by the point’s own model total.
- **Local density:** assumes $\rho_i/\rho_{DM} = \Omega_i/\Omega_{DM}$. A large departure requires an astrophysical model absent here.
- **Mediator and interference:** minimal TP-D has one diagonal SM-Higgs exchange amplitude per component. There is no second amplitude or adjustable sign, hence no tree-level blind spot. Higgs–radial or kinetic-mixing amplitudes belong to a modified TP-G benchmark and cannot be invoked retroactively.
- **Isospin and normalization:** Higgs exchange is approximately isospin conserving, with $f_N = 0.30$, and

$$\sigma_{iN}^{SI} = \frac{\lambda_{Hi}^2 f_N^2 \mu_{iN}^2 m_N^2}{4\pi m_h^4 m_i^2}.$$

Reasonable hadronic uncertainty cannot remove a minimum factor-37 excess.

- **Momentum dependence:** $m_h \gg q_{xenon}$, so the contact, momentum-independent SI mapping is valid.
- **Confidence convention:** LZ’s power-constrained 90% C.L. upper limit is applied.
- **Halo convention:** the comparison inherits LZ’s standard WIMP halo assumptions. The result is conditional on those assumptions, but the margins are not marginal.
- **Multicomponent likelihood:** a component that independently exceeds the rescaled single-component limit is sufficient for conservative exclusion because the other component adds a nonnegative event rate. A combined likelihood is needed only when every component is individually below its limit.

11.110 Analytic reason and next reduced search

The failed points obtain freeze-out depletion using Higgs portals large enough to produce xenon rates 37–312 times above LZ. Away from poles and thresholds, both Higgs-mediated annihilation and direct detection scale as λ_{Hi}^2 ; simply lowering a portal raises the relic abundance and does not generically lower $\xi_i \sigma_i$. Conversion $B\bar{B} \leftrightarrow A\bar{A}$ conserves total dark-particle number and is not an independent depletion mechanism.

The next justified reduced directions are:

1. $m_i \simeq m_h/2$, requiring finite-width thermal averaging before any resonance scan.
2. Semi-annihilation dominance with a near-maximal vacuum-safe cubic and a small Higgs portal, with exact vacuum and finite-energy unitarity filters.
3. Only after TP-G repair: secluded annihilation into lighter dark vectors/radials.

4. Only as a costed modified model: Higgs–radial interference blind spots.

No new broad random scan is authorized by this result.

11.111 Claim update

- **Strengthened, verified model-dependent result:** all four previously “relic compatible” random-scan points are excluded under the standard SI/local-density mapping.
- **Invalidated claim:** these four points cannot be called viable numerical benchmarks.
- **Preserved exact result:** TP-D’s kernel selection rules and forbidden $B \rightarrow A + h$ amplitude are unaffected.
- **Open:** whether any resonance, semi-annihilation-dominated, or repaired secluded TP-G region is simultaneously relic-, direct-detection-, vacuum-, unitarity-, and RG-consistent.

11.112 Exact unresolved dependency order

1. Implement finite-width thermal averaging and run the smallest Higgs-pole scan.
2. Bound the semi-annihilation escape direction analytically and with a minimal targeted grid.
3. Decide whether any TP-D point survives all phenomenology gates.
4. Continue gauge/Goldstone/transverse-vector finite-energy unitarity.
5. Complete parent gauge/scalar/kinetic-mixing RG matching, anomalies, perturbativity, and EFT cutoff.
6. Complete loop-improved vacuum/metastability and applicable cosmology.
7. Finish novelty, parameter-reducing observable, hostile-referee, and final equation/claim regression audits.

Machine-readable results: `audits/tpd_direct_detection_gate.json`. Reproduction code: `code/tpd_direct_detection_gate`

Regression status: **29/29 tests pass**.

11.113 S10 TP checkpoint 15 — point-matched TP-G finite-energy gauge gate

Completion-protocol checkpoint: C14

Verdict: **CONDITIONAL PASS**

Scope: frozen Higgs-pole point matched to the charge-conjugation-protected zero-kinetic-mixing TP-G parent, through 20 TeV.

11.114 Point-specific matching

For each m/E Abelian sector,

$$f = 1 \text{ TeV}, \quad g = 0.30, \quad Q_\Phi = 3, \quad M_X = 900 \text{ GeV},$$

$$\lambda_\Phi = 0.20, \quad m_\rho = 632.456 \text{ GeV}, \quad \lambda_{S\Phi} = 0.$$

The dark masses are 62.0 and 62.2 GeV, with $\lambda_S = 0.10$ and

$$\kappa = \frac{\sqrt{2}\mu_S}{3f} = 4.71405 \times 10^{-4}$$

for $\mu_S = 1 \text{ GeV}$. The two sectors factorize in the gauge calculation; the tiny A – B and Higgs portals remain in the already-tested scalar block.

11.115 Methods

- Physical massive-vector helicity amplitudes were constructed for longitudinal and transverse states.
- The normalized neutral $J = 0$ basis was $\{X_L X_L / \sqrt{2}, \rho\rho / \sqrt{2}, SS^\dagger\}$.
- The mixed charged block $XS \leftrightarrow \rho S$ was projected for $J = 0, 1, 2, 3, 4$.
- Other pure-vector and charged helicity entries were bounded for every J using $|d_{mm'}^J| \leq 1$.
- Identical-state factors, thresholds, radial/vector exchange, contact diagrams, and angular convergence were included.
- The physical $J = 1$ vector resonance was width-resummed using $X \rightarrow SS^\dagger$.
- Scalar-QED Ward contractions and fixed-angle longitudinal/Goldstone equivalence were checked.

11.116 Controlled results

Diagnostic	Result
Largest neutral longitudinal $J = 0$ eigenvalue magnitude	0.190989
Strongest non-LLLL pure-vector all- J entry bound	0.031975
Largest charged-block eigenvalue in the clean regions, $J = 0 \dots 4$	0.0613163
Strongest charged all- J entry bound in the clean regions	0.0613017
Scalar-QED Ward relative residual	$< 8.5 \times 10^{-16}$
Compton Ward relative residual	$< 3.0 \times 10^{-16}$
Longitudinal/Goldstone mismatch at 20 TeV, fixed angle	0.28%–0.82%
$X_m \rightarrow AA^\dagger$ width	0.52193 GeV
$X_E \rightarrow BB^\dagger$ width	0.52183 GeV
$\kappa/(16\pi)$ amplitude scale	9.38×10^{-6}

The 80/160/320-node values of the most forward-sensitive normalized $X_L X_L \rightarrow \rho\rho$ wave at 20 TeV are -0.179808572 , -0.179808667 , and -0.179808667 . The largest controlled eigenvalue therefore also satisfies the stronger comfort criterion $|a_J| < 1/4$.

11.117 Resonance and exact sequential-pole diagnosis

At $\sqrt{s} = M_X$, the width-resummed elastic $SS^\dagger$ $J = 1$ wave has

$$a_1 \simeq -4.16 \times 10^{-4} + i,$$

and lies on the elastic Argand circle to 1.8×10^{-7} . This is a physical resonance, not a tree-unitarity violation.

For $X(M) + S(m) \rightarrow X(M) + S(m)$, the exact exchanged-scalar condition is

$$u - m^2 = M^2 - 2E_X E_S - 2p^2 \cos \theta = 0.$$

Its angular solution first enters and finally leaves the physical interval at

$$\sqrt{s}_{\text{low}} = \sqrt{2M^2 + m^2}, \quad \sqrt{s}_{\text{high}} = \frac{M^2 - m^2}{m}.$$

The ordering $\sqrt{s}_{\text{high}} > \sqrt{s}_{\text{low}}$ holds if and only if $M > 2m$, exactly the decay condition $X \rightarrow SS^\dagger$. Numerically the contaminated intervals are:

- 1.2743–13.0025 TeV in the m sector;
- 1.2743–12.9603 TeV in the E sector.

Thus X is unstable and is not a conventional asymptotic external state. This interval is assigned neither PASS nor FAIL and is not numerically regularized. The ordinary two-body waves in the clean regions below and above it remain below $1/2$; high-energy helicity cancellations are included in the all- J bounds.

11.118 Verdict

CONDITIONAL PASS. No properly normalized, nonsingular stable-state channel shows a genuine perturbative-unitarity or Ward-identity failure through 20 TeV. The largest controlled formal gauge eigenvalue is $0.191 < 1/2$. The XS interval is unscored. An unconditional parent full-S-matrix PASS would require an inclusive calculation using stable asymptotic states or a consistently defined unstable-particle scheme.

Nonzero Abelian kinetic mixing is outside this protected benchmark and is not a loophole needed to save it.

Reproduction: `code/tpd_parent_gauge_finite_energy.py`; results: `audits/tpd_parent_gauge_finite_energy.json`.

11.119 S11 TP checkpoint 16 — Higgs-pole thermal and experimental certification

Completion-protocol checkpoint: C15

Verdict: **CONDITIONAL PHENOMENOLOGICAL PASS**

11.120 Frozen parameter vector

Parameter	Value
m_A, m_B	62.0, 62.2 GeV
$\lambda_{HA}, \lambda_{HB}$	$5.5393817 \times 10^{-4}, 4.9170872 \times 10^{-4}$
μ_A, μ_B	1, 1 GeV
$\lambda_A, \lambda_B, \lambda_{AB}$	$0.1, 0.1, 10^{-5}$

11.121 Thermal certification

The coupled Boltzmann system was recomputed with direct finite-width Gondolo–Gelmini thermal averages, the off-pole continuum, temperature-dependent $g_\rho(T)$ and $g_s(T)$, and the $d \ln g_s / d \ln T$ correction. Scaled Bessel functions remove the former large- x numerical underflow.

Quantity	Result
$\Omega_A h^2$	0.0592356
$\Omega_B h^2$	0.0596303
$\Omega_{\text{tot}} h^2$	0.118866
ξ_A, ξ_B within the predicted abundance	0.498340, 0.501660
110-to-150-point rate-table shift	5.29×10^{-5}
g_ρ, g_s at $T = 3.1$ GeV	84.5747, 84.1187
EOS $\pm 5\%$ envelope for $\Omega_{\text{tot}} h^2$	0.11593–0.12205

The nominal result is 0.945% below the reference 0.120. This is a viable numerical benchmark at the present leading-order precision, not a precision relic prediction.

11.122 Refined direct detection and collider checks

Fractions below use $\Omega_{DM} h^2 = 0.1200$, as required for abundance-weighted comparison.

Component	ξ_i	raw σ_{iN}^{SI} [cm ²]	effective [cm ²]	LZ limit [cm ²]	R_i
<i>A</i>	0.493630	6.8030×10^{-49}	3.3582×10^{-49}	2.5060×10^{-48}	0.13401
<i>B</i>	0.496919	5.3265×10^{-49}	2.6468×10^{-49}	2.5087×10^{-48}	0.10551

The masses differ by only 0.2 GeV: their xenon recoil-scale proxy differs by 0.43%, so at exclusion-level precision the spectra are indistinguishable and the rate ratios add to $R_{\text{combined}} = 0.23952 < 1$.

The predicted invisible-Higgs branching fraction is 1.6881×10^{-4} , versus the ATLAS observed 95% C.L. bound 0.107. In minimal TP-D there is no Higgs mixing because both dark scalars have zero vevs. In the protected TP-G slice, kinetic mixing and Higgs–parent-radial portals vanish at matching; switching either on requires a separate collider analysis.

11.123 Classification and limitation

The point survives the current relic, direct-detection, and collider gate at leading order. Dominant limitations are the approximate QCD equation of state, higher-order energy-dependent virtual-Higgs width, and use of the published LZ exclusion curve rather than a two-component detector likelihood. None is numerically close to reversing the present factor-4.17 rate margin.

Reproduction: `code/tpd_pole_thermal_certification.py`; results: `audits/tpd_pole_thermal_certification.json`.

11.124 S12 TP checkpoint 17 — cubic processes and point-matched metastability

Completion-protocol checkpoint: C16

Verdict: **CONDITIONAL PASS**

11.125 Cubic-process gate

For both $m_A = 62.0$ GeV and $m_B = 62.2$ GeV, the zero-temperature semi-annihilation channel $SS \rightarrow S^\dagger h$ is exactly closed because $m_S < m_h$.

A deliberately conservative five-point amplitude envelope,

$$|\mathcal{M}_{3 \rightarrow 2}| \leq 100 \frac{|\mu_S \lambda_S|}{m_S^2},$$

gives at $x = 20$

$$\max_S \frac{n_{S,eq}^2 \langle \sigma v^2 \rangle}{H} = 2.19 \times 10^{-9}.$$

Thus cubic-induced $3 \rightarrow 2$ reactions cannot change the certified freeze-out result at relevant accuracy; no enlarged Boltzmann system is justified.

11.126 Point-matched one-loop running

The parent running was repeated with the actual pole-point low-energy quartics and portals, and with

$$g_m = g_E = 0.30, \quad \lambda_{\Phi_m} = \lambda_{\Phi_E} = 0.20, \quad \kappa_m = \kappa_E = 4.71405 \times 10^{-4}$$

at the 1 TeV matching scale.

Diagnostic	Scale/result
First failure of the sufficient BFB certificate	3.36215×10^8 GeV
High-energy scalar-unitarity boundary	5.28504×10^{17} GeV
First 4π coupling boundary	1.03367×10^{18} GeV
λ_H at scalar-unitarity cutoff	-0.0297107

The instability is the familiar Higgs-direction loss of absolute stability, not a cubic-driven dark direction.

11.127 Metastability envelope

Using the leading-log quartic bounce envelope,

$$S_4 \simeq \frac{8\pi^2}{3|\lambda_H|} = 885.8,$$

and the conservative four-volume prefactor capped at the first scalar partial-wave unitarity boundary, 5.285×10^{17} GeV, gives $\ln P_{\text{decay}} < -337$. The electroweak vacuum is therefore cosmologically long-lived in this approximation. No bounce claim is extrapolated into the interval between the scalar-unitarity and 4π boundaries.

This is not a full multi-field loop-improved bounce calculation. Finite threshold corrections, dimensionful running in the parent theory, two-loop evolution, gauge dependence of the effective potential, and a numerical multi-field bounce remain open precision tasks. Accordingly, the scientifically valid result is a **conditional metastability pass**, not an absolute-stability claim.

Reproduction: `code/tpd_cubic_metastability.py`; results: `audits/tpd_cubic_metastability.json`.

11.128 S13 TP checkpoint 18 — point-specific parent cosmology

Completion-protocol checkpoint: C17

Verdict: PASS at leading order

11.129 Extra parent states

At the matched pole point, $f_m = f_E = 1$ TeV, $g_m = g_E = 0.30$, $M_X = 900$ GeV, and $m_\rho = 632.456$ GeV. Unlike the already rejected B0 benchmark, both parent sectors have open prompt decays.

Sector	Decay	Width [GeV]	Lifetime [s]	$\Gamma/H(T = M)$
m vector	$X_m \rightarrow AA^\dagger$	0.521926	1.26×10^{-24}	4.59×10^{11}
E vector	$X_E \rightarrow BB^\dagger$	0.521828	1.26×10^{-24}	4.59×10^{11}
m radial	$\rho_m \rightarrow AAA + \text{c.c.}$	3.9212×10^{-8}	1.68×10^{-17}	6.98×10^4
E radial	$\rho_E \rightarrow BBB + \text{c.c.}$	3.9144×10^{-8}	1.68×10^{-17}	6.97×10^4

The radial widths use the exact massive three-body phase-space integral. From $\kappa\Phi^\dagger S^3 + \text{h.c.}$ and $\mu = 3\kappa f/\sqrt{2}$, each charge-channel amplitude is $|\mathcal{M}| = 2\mu/f = 0.002$. These lifetimes exclude late entropy injection and BBN disruption by many orders of magnitude.

11.130 Late-time stable-state checks

The zero-velocity Higgs-mediated annihilation rates are 1.47×10^{-28} and $2.49 \times 10^{-28} \text{ cm}^3 \text{ s}^{-1}$. Including the squared component fractions and conservatively setting the deposition efficiency to one gives

$$p_{\text{ann}} = 1.56 \times 10^{-30} \text{ mcm}^3 \text{ s}^{-1} \text{ GeV}^{-1},$$

or 0.00489 of the Planck 2018 95% C.L. limit $3.2 \times 10^{-28} \text{ cm}^3 \text{ s}^{-1} \text{ GeV}^{-1}$.

A deliberately loose low-velocity self-scattering amplitude envelope gives $\sigma/m < 4.6 \times 10^{-12} \text{ cm}^2 \text{ g}^{-1}$ for either component. This is phenomenologically negligible.

Both parent Goldstones are eaten. The broken gauge factors produce local strings, not domain walls; their TeV-scale tension estimate is only $G\mu \sim 1.1 \times 10^{-30}$. Thus there is no dark-radiation or relevant defect constraint in this matched slice.

11.131 Classification

Verified model-dependent result: the pole benchmark has no unintended long-lived parent state and passes the applicable leading-order BBN, late-entropy, CMB-injection, self-interaction, dark-radiation, and defect checks. A nonstandard cosmological history was neither assumed nor needed.

Reproduction: `code/tpd_parent_extra_state_cosmology.py`; results: `audits/tpd_parent_extra_state_cosmology.json`.

11.132 S14 TP checkpoint 19 — adversarial novelty and prior-art gate

Completion-protocol checkpoint: C18

Verdict: **FAIL** for broad novelty; narrow formulation claim remains unproven

11.133 Closest established mechanisms

The comparison was deliberately limited to primary papers that decide the gate:

Claimed ingredient	Prior-art result	Consequence for TP-D
Z_N scalar semi-annihilation and cross-sector reactions	Bélanger et al., arXiv:1202.2962	established
local $U(1) \rightarrow Z_3$ with a charge-three dark Higgs, vector, and radial	Ko and Tang, arXiv:1402.6449	TP-G mechanism established
several stable scalar species under discrete symmetry	Yaguna and Zapata, arXiv:1911.05515	multicomponent stabilization established
product residual discrete gauge symmetry organizing dark states	Borah, Ma, and Nanda, arXiv:2212.11847	product-remnant mechanism established

A targeted search for the exact two-singlet $Z_3 \times Z_3$ operator basis and the particular projection/kernel language did not produce a closer indexed match. Absence from a targeted search is not evidence of novelty and is not promoted to a claim.

11.134 Adversarial claim classification

Statement	Classification
The kernel forbids the enumerated cross operators	exact mathematical/model result
The forbidden surface is radiatively protected	exact symmetry result
TP-D removes 12 physical continuous coefficients relative to the stated generic projected- Z_3 comparator	verified model-dependent result
$B \rightarrow A + X_0$ vanishes in exact TP-D	exact selection rule
Product discrete dark matter, semi-annihilation, or residual local Z_3 is new	excluded/failed claim
The names “matter-like” and “energy-like” encode new physics	unsupported conjecture
The projection/kernel organization is itself literature-new	unresolved; not demonstrated
TP-D is a new fundamental theory rather than a conventional product-discrete EFT	excluded/failed claim at current evidence

11.135 Gate result

The defensible scientific contribution is a fully audited, kernel-sensitive $Z_3 \times Z_3$ EFT with a viable pole benchmark and explicit null selection rules. It is not a demonstrated novel symmetry mechanism. Any publication claim must use “specific construction/audit” or “known ingredients combined,” not “new class of dark matter symmetry,” unless a substantially more exhaustive professional literature review establishes otherwise.

Primary comparison set: arXiv:1202.2962, 1402.6449, 1911.05515, and 2212.11847. Search checked 19 September 2026.

11.136 S15 TP checkpoint 20 — parameter reduction and falsifiability

Completion-protocol checkpoint: C19

Verdict: **WEAK PASS** for exact null relations; **FAIL** for a unique nonzero TP-D relation

11.137 Exact minimal TP-D

The complete operator audit removes twelve physical continuous coefficients relative to the generic two-field projected- Z_3 comparator. This produces a correlated family of parameter-free observable nulls,

$$\mathcal{M}(B \rightarrow A + X_0) = 0,$$

and analogous kernel-changing scattering amplitudes, whenever X_0 is kernel neutral and the exact symmetry is unbroken. These are genuinely parameter-reducing and falsifiable: one verified nonzero kernel-changing amplitude rejects exact minimal TP-D.

This satisfies the minimal “at least one falsifiable relation” criterion, but only weakly. A generic comparator can tune the same amplitudes to zero, so a null observation cannot positively identify TP-D. Identification would require both charge eigenstates, measured nonzero diagonal interactions, and several independently open but absent kernel-changing channels.

11.138 Controlled soft extension

If the only breaking is a dimension-two $A^\dagger B$ mass term while the hard off-diagonal Higgs portal remains zero, the mass-basis Higgs couplings obey

$$2G_{12} \cos 2\theta = (G_{22} - G_{11}) \sin 2\theta.$$

The generic projected- Z_3 theory instead has

$$2G_{12} \cos 2\theta - (G_{22} - G_{11}) \sin 2\theta = 2v\eta_{HAB}.$$

The identity is basis-covariant once another interaction—such as the parent gauge current—operationally identifies the charge basis. It was checked at 1000 random points to an absolute residual below 6.9×10^{-13} GeV.

This is a real parameter-reducing nonzero sum rule, but it belongs to TP-Bsoft, not the exact viable TP-D pole benchmark, and its two-state alignment algebra is standard. It therefore cannot be used to rescue a strong TP-D novelty claim.

11.139 Final classification

- **Exact result:** TP-D has parameter-free kernel-changing null amplitudes.
- **Falsifiability:** PASS—nonzero kernel-changing transitions falsify the exact model.
- **Positive identifiability:** unresolved and experimentally difficult.
- **Unique nonzero TP-D sum rule:** absent.
- **Soft-extension relation:** verified model-dependent result, not established novelty.

Reproduction: `code/tpd_soft_breaking_sum_rule.py`; results: `audits/tpd_soft_breaking_sum_rule.json`.

11.140 S16 TP checkpoint 21 — hostile-referee audit

Completion-protocol checkpoint: C20

Verdict: **CONDITIONAL SURVIVAL AS A CONVENTIONAL EFT BENCHMARK**

11.141 Referee decision

No fatal algebraic contradiction, stable-state perturbative-unitarity violation, anomaly, relic-density failure, direct-detection exclusion, or applicable cosmological exclusion was found for the Higgs-pole point. The benchmark survives, but several stronger interpretations fail.

11.142 Findings by severity

11.142.1 Corrected during review

The first metastability envelope had used the later 4π coupling boundary as its prefactor scale although high-energy scalar partial-wave unitarity is lost earlier. The calculation is now capped at 5.285×10^{17} GeV, the first scalar-unitarity boundary. The corrected values are $S_4 = 885.8$ and $\ln P < -337$; the qualitative conditional metastability verdict is unchanged.

11.142.2 Major limitations

1. The relic result is a leading-order numerical benchmark near a narrow Higgs pole. Its 0.95% target offset is inside the tested EOS envelope, but precision thermal-QCD input and higher-order virtual-Higgs treatment are absent.
2. Direct detection uses abundance-rescaled published LZ limits and a justified nearly-degenerate rate sum, not a detector-level two-component likelihood. The combined ratio 0.2395 leaves a sufficient present margin but is not a global fit.
3. Parent RG evolution uses a common 1 TeV step threshold. Finite matching, two-loop running, and parent dimensionful beta functions are not complete. The declared leading perturbative EFT ceiling is therefore $\Lambda_{EFT} \lesssim 5.3 \times 10^{17}$ GeV, not a UV-completion scale.
4. The parent vector is unstable. The physical XS sequential interval is correctly unscored; an unconditional full-S-matrix statement would require stable asymptotic states or a consistent unstable-particle inclusive scheme.
5. Vacuum longevity is supported by a one-loop leading-log Higgs-direction envelope, not a gauge-controlled numerical multi-field bounce.
6. Zero kinetic mixing is an extra charge-conjugation-protected slice. It is not forced by $U(1)_m \times U(1)_E$ alone.
7. The exact selection rule is formally falsifiable but experimentally inaccessible in the frozen pole point without an additional production/tagging structure. It is not positive evidence for TP-D.

11.142.3 Claims rejected

- broad symmetry or dark-matter-mechanism novelty;
- a unique nonzero TP-D observable relation;
- absolute stability to the Planck scale;
- an unconditional PASS assigned to the XS pole interval;
- explanations of generations, gravity, constants, quantum gravity, or unification.

11.143 Surviving scientific statement

TP-D has one conditionally viable two-component scalar Higgs-pole benchmark with exact kernel selection rules. Its scalar, electroweak-vector, controlled parent-gauge, leading one-loop RG/metastability, collider-prefilter, direct-detection, and necessary cosmology checks are mutually consistent within explicitly recorded approximations. It remains a conventional product-discrete EFT, not a demonstrated fundamental theory.

11.144 Exact unresolved work

- finite one-loop matching at the split parent thresholds and two-loop uncertainty;
- parent dimensionful RG evolution and a gauge-controlled multi-field bounce;
- detector-level two-component xenon likelihood if precision exclusion is required;

- stable-state inclusive treatment if an unconditional parent-vector S-matrix claim is desired;
- a concrete production/tagging setup for the kernel nulls;
- professional exhaustive prior-art review if any narrow novelty claim is pursued.

11.145 S17 TP checkpoint 22 — final equation and claim regression

Completion-protocol checkpoint: C21

Verdict: **PASS WITH EXPLICIT CONDITIONALS**

11.146 Automated result

All 34 unit/regression tests pass. The cross-artifact freeze regression passes all 13 checks, including:

- preservation and exclusion status of the four original relic-compatible scan points;
- exact preservation of the P0 parameter vector;
- finite-width thermal abundance and numerical convergence;
- separate and combined abundance-weighted direct-detection ratios;
- electroweak and parent gauge blocks;
- explicit unscored treatment of both physical XS sequential-pole intervals;
- absence of any stable-state genuine unitarity violation;
- correct ordering of BFB, scalar-unitarity, and 4π boundaries;
- metastability evaluation capped at the scalar-unitarity boundary;
- prompt parent-state decays and conservative CMB safety;
- claim-ledger continuity through M21.

Machine record: `audits/tpd_final_regression.json`.

11.147 Authoritative continuation state

The promoted state is `TP_D_canonical_definition_v0.3.md`. P0 is a **conditionally viable numerical benchmark** of a conventional $Z_3 \times Z_3$ scalar EFT. The parent gauge result is conditional only because unstable-vector external states do not define a conventional asymptotic S-matrix across the physical sequential interval; that interval is not a perturbative-unitarity failure.

The original four scan points remain excluded. Broad novelty and a unique nonzero TP-D sum rule remain failed claims. Absolute stability, experimental confirmation, generation counting, gravity, fundamental constants, quantum gravity, and unification are not claimed.

11.148 Exact unresolved tasks

1. finite one-loop matching at the split 0.63/0.90 TeV thresholds;
2. two-loop uncertainty and parent dimensionful running;
3. gauge-controlled numerical multi-field bounce;
4. detector-level two-component xenon likelihood only if precision exclusion is needed;
5. stable-state inclusive calculation only if an unconditional parent-vector S-matrix claim is pursued;
6. production/tagging construction for experimentally actionable kernel nulls;
7. exhaustive professional prior-art review only if a narrow novelty claim is pursued.

No additional broad scan is scientifically justified before one of these precision objectives is explicitly selected.